Sources of Labor Market Power*

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Abstract

Wages are often set below the marginal productivity of labor, reflecting deviations from perfect competition. We distinguish between two sources of such markdowns: classical labor market power, arising from imperfect substitution between labor markets, and search labor market power, linked to frictions in matching workers to jobs. We present a unified theoretical framework that incorporates both mechanisms and develop empirical tests based on the cross-sectional relationship between posted vacancies and wages, as well as their differential responses to demand shocks. Using granular job posting data and high-frequency monetary policy surprises, we find that search labor market power is the primary driver of cross-firm variation in wage markdowns and hiring responses. Our results suggest that markdowns may be partly efficient.

JEL Codes: E24, E43, E52, J23, J31, J42, J63

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1 Introduction

Wages in the U.S. are on average 30% below the marginal productivity of labor (Hershbein et al., 2022), indicating significant monopsony or labor market power. But where do these markdowns come from? What are their macroeconomic implications? And are they necessarily inefficient?

The theoretical literature suggests two possible sources of wage markdowns. "Classical" labor market power (Berger et al., 2022; Bhaskar et al., 2002; Robinson, 1969) stems from imperfect substitution across labor markets. In this setting, firms mark down wages and inefficiently curtail employment because workers are unable or unwilling to move between markets, e.g. due to specific skill requirements, geographical constraints, or individual preferences. In contrast, "search" labor market power (Jarosch et al., 2019; Burdett and Mortensen, 1998; Moen, 1997; Pissarides, 1985) stems from uncertainty in the matching process between workers and vacancies. Firms whose vacancies are easier to fill are able to mark down wages because they internalize that workers value more certain employment. As a result, search monopsony can lead to markdowns that may be efficient or, even if inefficient, could be associated with overemployment.[1]

In this paper, we propose a simple framework that allows for classical and search labor market power. The model builds on commonly used "classical" labor market power frameworks by enriching them with a simplified search friction. We first derive model predictions for the case when wage markdowns are due to classical monopsony alone, and then we consider how these predictions change when allowing for search monopsony. We test these predictions empirically by studying the relationship between the number of posted vacancies and wages and the responses of vacancies and wages to demand shocks, measured by monetary policy surprises, to ascertain the relative importance of either.

We find that classical monopsony models cannot rationalize neither the relationship between equilibrium vacancies and wages, nor the response of labor market variables to monetary policy. A model of classical monopsony would deliver a positive correlation between equilibrium vacancies and wages. Instead, search labor market power generates the observed negative relationship between vacancies and wages. This finding suggests that the variation in equilibrium markdowns across firms is mostly explained by heterogeneity in search rather than classical labor market power. Search labor market power is also crucial for the response of labor market variables to changes in demand, rationalizing the

[1] In standard Diamond-Mortensen-Pissarides models, the relative bargaining power of workers and firms determines a condition for efficiency originally derived by Hosios (1990) which may imply underemployment (workers' bargaining power is too large), overemployment (firms' bargaining power is too large) or optimal employment (markdowns fairly compensate firms). Examples of all three possibilities can also be found in models beyond the standard Diamond-Mortensen-Pissarides model, for instance, overemployment in Rudanko 2023, optimal employment where markdowns are efficient in Moen 1997, and optimal employment independent of the level of the markdown in Jarosch et al. 2019.

increased response of vacancies and muted response of wages to these shocks.

In the model with only "classical" labor market power, each firm has its own labor market, and workers allocate their search across firm-specific labor markets. The substitution between markets is imperfect – leading to the emergence of classical labor market power. This follows standard "classical" labor market power models from the recent literature, such as Berger et al. (2022), and extends the framework to many other common labor supply formulations.

This baseline model with "classical" labor market power only is then extended to include search frictions that make it uncertain for households whether they will find a job. In this case, households take into account both offered wages and the probability of finding a job in each market. Firms can thus offer value to workers in two ways: by raising wages or by offering more certain employment. The ability to generate more certain employment determines what we call "search" labor market power. Search labor market power is introduced in a simplified framework loosely based on the static version of a directed search model by Moen (1997). Unlike the textbook version of directed search models, in our model the substitution between individual labor markets is not perfect, allowing for firms to have both, either, or neither classical and search labor market power.

We first construct a test based on equilibrium wages and vacancies derived from firm decisions. Taking the search decision of workers (i.e., the labor supply) as given, firms choose wages and the number of vacancies to maximize profits. Firms with greater classical market power face an inelastic labor supply because workers cannot easily switch away from their labor market. Firms with classical monopsony internalize that raising wages affects their wage bill, and thus post lower wages and fewer vacancies, taking advantage that workers are less sensitive to both vacancies and wages. In contrast, firms with more search market power face workers that are highly reactive to the number of vacancies posted. This makes it relatively easier to hire an additional worker by posting an additional vacancy rather than by raising posted wages. In equilibrium, these firms post relatively more vacancies and pay lower wages. This means that comparing a firm's vacancy-level wages and the relative prevalence of its vacancies in the labor market where the vacancy is posted can be a valid test for the most relevant source of monopsony: if wages are positively related to the share of vacancies posted by the firm, that suggests classical monopsony is more important, otherwise search monopsony is more important.

Because equilibrium wages and vacancies may be empirically hard to pin down and suffer from confounding factors, we also derive predictions from responses to exogenous demand shocks. Under classical monopsony power, firms with more market power will not react differentially to demand shocks, unless their wages respond differentially. In contrast, under search labor market power, firms with more market power post relatively more vacancies in response to a demand shock, even if their wages do not adjust differentially. To derive a second test for which source of market power is dominant, we map the predictions of the model to responses that depend on firms' equilibrium wages and vacancies, because we do not observe directly the degree of either classical or search monopsony. The test states that if firms only vary in the extent of classical monopsony alone and wages do not respond differentially, we should expect no heterogeneity in the response of vacancies to monetary policy across firms regardless of their equilibrium wages or vacancies. On the contrary, if search labor market power is very prevalent, vacancies of firms with lower wages and higher vacancies in equilibrium should respond more to demand shocks. These very different predictions allow us to test the relative importance of classical and search monopsony for the dynamics of labor market outcomes.

We take these two tests to data on the near universe of online job postings provided by Lightcast (formerly Burning Glass Technologies). Lightcast data cover 250 million online job vacancy postings that correlate strongly with firm-level changes in net employment. Details for each vacancy include information on the firm, location, posted date, job requirements, and offered wage, among others. The highly disaggregated data allow us to construct a firm-local labor market level dataset, effectively allowing for establishment-level data. This dataset is perfectly suited for our purposes, as it focuses on postings by firms as opposed to equilibrium outcomes for employment that are influenced by the search decisions of households. Moreover, the observables in this dataset correspond perfectly to our modeling framework: in the model we assume that search is directed with wages and vacancy postings being observable by the workers, and our dataset exactly contains the information that is available to workers at the time of making search decisions.

We use unexpected high-frequency monetary policy shocks around Federal Open Market Committee (FOMC) meetings from Jarociński and Karadi (2020) to identify exogenous demand shocks. Central banks implement changes in monetary policy, such as adjustments in interest rates or quantitative easing, with the explicit goal of influencing aggregate demand. Using the unanticipated part of the monetary policy decision allows us to treat them as exogenous variations of aggregate demand. By analyzing vacancies and wages in response to these shocks, we can infer the effects of demand-side disturbances while minimizing concerns about reverse causality or omitted variable biases.

The empirical counterparts of our model-derived tests offer evidence in favor of the large heterogeneity of search labor market power, and reject important heterogeneity in classical labor market power. We find a negative relationship between equilibrium levels of vacancies and wages and an amplification of the effect of monetary policy on vacancy postings for firms with higher equilibrium vacancies and lower equilibrium wages. Both findings reject the hypothesis that heterogeneity in classical labor market power is a key feature of the data. The effects are significant and large. Quantitatively, a firm at the 50th percentile of vacancy share distribution increases its labor demand by $\approx 7\%$ in response to a 10 basis point surprise monetary loosening while a firm at the 95th percentile of the vacancy share distribution increases labor demand by $\approx 9\%$. Moreover, the effect of monetary policy shocks on firms with market power is persistent, with effects remaining economically large and statistically significant for at least six quarters. We also find that these results are driven by low-skilled vacancies, which a priori could be more sensitive to both classical and search monopsony, and thus this finding reinforces the idea that search monopsony is critical to understand the dynamic responses of vacancies to demand shocks. Finally, we show that our findings for vacancies also extend to net employment by merging Lighcast with Compustat data. Search monopsony drives the response of not only labor demand but also equilibrium labor.

These findings have important implications for the transmission of monetary policy. A common view is that labor market power dampens the response of employment to policy shocks, as monopsonistic firms face a less elastic labor supply and therefore adjust hiring less. However, our results challenge this view. When labor market power arises from search frictions rather than imperfect substitution across markets, firms with greater labor market power respond more aggressively, not less aggressively, to demand shocks. In this case, search frictions amplify rather than attenuate the effects of monetary policy on labor market outcomes. Consistently, labor market power flattens the wage Phillips curve, as shown in Burya et al. (2023), and can explain the wageless recovery from the great financial crisis at the same time when employment was rising.

To isolate the effect of labor market power on wages and vacancies, it is crucial to rule out confounding firm-level differences, such as productivity, size, or regional characteristics. We therefore test whether wages and vacancies respond differentially to labor market power, both with and without controls. Without controls, we observe significant heterogeneity in wage responses, suggesting that firms with greater labor market power also differ systematically along other dimensions, such as being located in more productive regions or employing different types of workers. However, once we include firm, industry by year, commuting zone by year, and occupation by year fixed effects, and residualize posted wages at the occupationCZtime level, the heterogeneity in wage responses disappears and the negative wagevacancy slope remains. This pattern is consistent with our main interpretation that results are not driven by other firm-level differences but reflect the effects of labor market power. In addition, we redefine markets as CZ by industry, use vacancy shares instead of levels, limit the sample to postings with reported wages, remove duplicates, winsorize heavy posters, apply alternative Lightcast filters, and repeat the results with alternative monetary policy shocks, and we continue to find a differential vacancy response without a corresponding differential wage response.

Our paper contributes to several recent branches of the literature. Most notably, our results are important for the studies of labor market power, such as Berger et al. (2022); Jarosch et al. (2019); Hershbein et al. (2022); Azar et al. (2019a,b,c, 2020, 2022); Benmelech et al. (2022); Rudanko (2023); Burdett and Mortensen (1998); Pissarides (1985). However, the literature has neglected to consider the relative importance of classical and search labor market power in a unified framework. A notable recent exception is contemporaneous work by Berger et al. (2024). Using a unified framework and administrative employment data from Norway, the authors decompose the equilibrium markdown into three components coming from classical monopsony, bargaining, and search frictions, to show that the latter is the most important driver. However, Berger et al. (2024) abstract from the ability of firms to manipulate matching intensities by changing the number of vacancies that they post, which means their framework cannot speak to vacancy postings, firm labor demand, and employment. We show that pure "classical" labor market power models are not able to account for the patterns we observe in vacancy data, and argue those can be reconciled if vacancies affect matching probabilities, a form of search labor market power.

The modeling approach in our paper builds upon the monopsony models of Berger et al. (2022) and directed search, such as in Moen (1997) and classic paper of Pissarides (1985).

Our approach offers some preliminary insight for the literature on hiring intensity (Davis et al., 2013; Wolthoff, 2018), aggregate matching elasticity (Lange and Papageorgiou, 2020), and the Beveridge curve (Burdett et al., 2001). In particular, cross-firm heterogeneity in the degree of search monopsony would rationalize an amplification of the response of hiring to productivity shocks as in Wolthoff (2018), the findings from Lange and Papageorgiou (2020) that the aggregate matching elasticity is procyclical, and both the flattening and outward shift of Beveridge curve, documented by Burdett et al. (2001).

Our paper also contributes to the literature on the effects of monetary policy on labor markets, in particular Romer and Romer (1989); Fornaro and Wolf (2021); Coglianese et al. (2021); Dolado et al. (2021); Coibion et al. (2017); Andersen et al. (2021); Jasova et al. (2021); Bartscher et al. (2021); Bergman et al. (2022). It offers novel results on the response of vacancies and wages to monetary policy, and finds that effects coming from search labor market power are quantitatively significant, persistent, and are different from the ones suggested by the common intuition based on classical labor market power. These effects are of critical interest for monetary policy makers that need to target labor market outcomes, such as unemployment, market tightness, or wage growth.

Finally, our paper relates to the literature that uses data from Lightcast, formerly known as Burning Glass Technologies (BGT). Lightcast is among the best established datasets for vacancy postings. Papers that specifically looked at labor market power using this dataset include Hershbein and Kahn (2018); Hazell et al. (2021); Hershbein et al. (2022); Azar et al. (2022). Those papers mostly focus on the equilibrium effects of labor market power, such as the levels of wages, and do not explore either the specific frictions behind the markdowns or the effects of monetary policy on labor markets.

The remainder of this paper is organized as follows: Section 2 lays out a search and matching model and formulates tests on the relative prevalence of search and classical monopsony. Section 3 introduces our data and Section 4 details our empirical approach and results. Finally, Section 5 concludes.

2 Theory

This section introduces a model with both classical and search labor market power. We divide it into two subsections. In the first subsection, we consider a baseline model with only classical labor market power from Berger et al. (2022).[2] From this model, we derive predictions and formulate a null hypothesis that we test in Section 4. In the second subsection, we add a simplistic directed search framework in the spirit of Moen (1997) and investigate how this addition alters the predictions of the baseline model that we again test in Section 4.

2.1 Model of Classical Labor Market Power

General Environment We consider a large class of models with only classical labor market power with Berger et al. (2022) as a recent example. The economy consists of a representative household and a unit mass of firms. Workers choose search effort in each of the labor markets to maximize aggregate earnings. Firms hire workers in firm-specific labor markets, and produce a differentiable and imperfectly substitutable good to maximize instantaneous profit. Firms are non-atomistic, so that each firm's labor market takes up a non-zero share within the corresponding local labor market.

Household Problem We assume the representative household maximizes a utility function that is separable in search effort and consumption, which allows us to focus on labor market decisions exclusively.[3] Aggregate effort (N_t) is determined in the utility-

[2] Berger et al. (2022) distinguishes between "classical" settings with atomistic firms and their setting of "oligopsony" labor market power. This paper does not focus on the distinction between atomistic firms and oligopsony, and classifies both as "classical", in the sense that both rely on imperfect substitution between markets, and compares them with "search" labor market power.

[3] We depart nominally from Berger et al. (2022) by denoting the household labor decision in terms of search effort which in this model will equal labor because there are no search frictions.

maximization problem by trading off the aggregate wage (W_t) and the disutility of labor. The household chooses how much to search (n_{jt}) in the labor market j and how much to search (n_{ijt}) at firm i within this market, taking as given wages (w_{ijt}), and aggregate search across all labor markets (N_t), according to the following maximization problem:

$$W_t N_t = \max_{n_{ijt}} \int_j \left(\sum_i n_{ijt} w_{ijt} \right) dj$$

$$\int_j \left(\frac{n_{jt}}{N_t} \right)^{\frac{\theta+1}{\theta}} dj = 1 \qquad \sum_i \left(\frac{n_{ijt}}{n_{jt}} \right)^{\frac{\eta+1}{\eta}} = 1$$

where parameters θ and η are elasticities of the substitution between and within markets respectively.

Optimal search in each firm/labor market is given by the first-order condition:

$$\frac{w_{ijt}}{W_t} = \left(\frac{n_{ijt}}{n_{jt}} \right)^{\frac{1}{\eta}} \left(\frac{n_{jt}}{N_t} \right)^{\frac{1}{\theta}} \tag{1}$$

This optimal search condition defines labor supply in this model and coincides with equation (4) in Berger et al. (2022) by further noting that search efforts equal labor in this model without search frictions.

Firms' Problem Firms maximize profits by choosing their wage,[4] taking the demand for their goods and the amount of search allocated to their labor market as given. They choose which wages to pay. Classical oligopsony models do not usually consider search frictions, so in this version of the model we assume that posting vacancies is relatively costless and thus vacancies will equal to the number of hired workers which also equals households' search effort.

The revenue of firm i in market j is given by $p_{ijt} y_{ijt}$ and depends on the demand for the firm's goods and its production function. For this partial equilibrium set-up, we abstract from specifying the demand function and represent it by $y_{ijt} = \mathcal{Y}_{ijt}$, where $\mathcal{Y}_{ijt}$ depends on the firm's relative price, aggregate production and, possibly, other parameters. Aggregate production itself is a function $f(l, I)$ of labor and possibly other inputs. The firm's costs is given by payroll $w_{ijt} l_{ijt}$.

The profit maximization problem for firm i in market j is:

[4] In Berger et al. (2022) firms choose hiring instead, but that problem is isomorphic to the one laid out here. We set the problem up in terms of a choice of wages because that is also how we set up the extended model with search in the next subsection.

$$
\begin{aligned}
&\max_{v_{ijt}, w_{ijt}} p_{ijt} y_{ijt} - w_{ijt} l_{ijt} - Q_{ijt} \\
&v_{ijt} = l_{ijt} = n_{ijt} \\
&\frac{w_{ijt}}{W_t} = \left(\frac{n_{ijt}}{n_{jt}}\right)^{\frac{1}{\eta}} \left(\frac{n_{jt}}{N_t}\right)^{\frac{1}{\theta}} \\
&y_{ijt} = \mathcal{Y}_{ijt} \\
&y_{ijt} = f(l_{ijt}, I_{ijt})
\end{aligned}
$$

The first constraint denotes that in this model without search frictions and with costless vacancy postings posted vacancies equal the number of new hires which equals the amount of search effort for firm i in market j; the second constraint denotes the optimal labor supply decision from the household problem; the third and fourth constraints denote the demand and the production function, respectively. Note that this problem allows for capital to be an additional input into production (through the parameter vector I_{ijt} and investment costs are included in Q_{it}). The only assumption made is that labor and capital spending are separable in the cost function, so that Q_{ijt} does not depend on labor.

The first-order conditions of the problem above are:

$$
v_{ijt} = l_{ijt} \ , \qquad w_{ijt} = \frac{\xi_i^w}{\xi_i^w + 1} m_{ijt}
$$

where $\xi^w = \frac{\partial l_{ijt}}{\partial w_{ijt}} \frac{w_{ijt}}{l_{ijt}}$ is the elasticity of labor supply with respect to wage, and $m_{ijt} = (p_{ijt} y_{ijt})'_l$ is the marginal revenue of labor. Note that the marginal revenue of labor depends on labor productivity (which is a function of I_{ijt}) and on the firm's markup in the product market (which is embedded in the product demand $\mathcal{Y}_{it}$). For labor market outcomes, it does not matter whether higher marginal revenue of labor comes from higher productivity or higher product market power, because these only influence the hiring decision through m_{ijt}. The equilibrium equation for wages above corresponds to equation (7) in Berger et al. (2022) except we have not assumed a particular functional form for the production function.

The elasticity of labor supply can be found from the household optimal search equation and is given by:

$$
\xi_i^w = \frac{1}{\gamma_{ijt}} \ , \quad \gamma_{ijt} = \frac{1}{\eta} + \lambda_{ijt} \left(\frac{1}{\theta} - \frac{1}{\eta}\right)
$$
$$
\text{where } \lambda_{ijt} = \frac{w_{ijt} l_{ijt}}{\sum_i w_{ijt} l_{ijt}}
$$

which correspond to the unnumbered equations below equation (7) in Berger et al. (2022).

γ_{ijt} is the inverse of the elasticity of labor supply and is going to be very important going forward. This parameter summarizes the extent of "classical oligopsony", so we are going to call it the "classical oligopsony parameter", with larger γ_{ijt} indicating greater classical oligopsony power. In this model, γ_{ijt} depends positively on the payroll share λ_{ijt}: firms that control a larger share of payroll have more classical oligopsony power.

Solving for the elasticities using the inverse labor supply equation, the first order conditions to the firm's problem above simplifies to:

$$v_{ijt} = l_{ijt} = n_{ijt} \ , \qquad w_{ijt} = \frac{1}{\gamma_{ijt}+1} m_{ijt} \tag{2}$$

Note that, as expected, greater classical oligopsony power (higher γ) is associated with more marked down wages. The markdown, or the extent to which firms pay wages below the marginal revenue of labor, is given by a familiar Lerner formula (Lerner, 1934):

$$\nu_{ijt} = \frac{1}{\gamma_{ijt}+1} \tag{3}$$

With the goal of bringing this model to data, we next derive the equilibrium relationship between the share of vacancies of a firm in a market ($s_{ijt} = v_{ijt}/\sum_i v_{ijt} = n_{ijt}/\sum_i n_{ijt}$) and its equilibrium wages. In this setting, there is essentially no distinction between vacancies and hiring, so the relationship between equilibrium vacancies and wages is trivial and is determined by the positive slope of the labor supply. To derive this relationship, we take a log-linearization of labor supply (equation 1) and the equilibrium conditions (equations 2) around the average firm in the average market, such that $n_{ijt} = n_{jt} = N_t, s_{ijt} = s, \gamma_{ijt} = \gamma$. We consider heterogeneity across firms in terms of marginal revenue from labor m_{ijt} and labor supply parameters η and θ, which are subsumed in γ_{ijt}, while keeping other firms' variables and economy-wide aggregates fixed. The following denotes the resulting expression for wages and the share of vacancies and for the covariance between wages and the share of vacancies in the cross-section of firms, where tildes denote departures from the average firm in the average market:[5]

$$\tilde{w}_{ijt} = \gamma \tilde{n}_{ijt} \ , \qquad \tilde{s}_{ijt} = \frac{1-s}{\gamma} \tilde{w}_{ijt} \ , \qquad \text{cov}(\tilde{w}_{ijt}, \tilde{s}_{ijt}) = (1-s) ma\text{var}(\tilde{n}) > 0$$

The relationship between equilibrium wages and vacancy shares reflects two effects: on the one hand, higher wages mean higher employment (and vacancies) due to a positive slope of the labor supply; while, on the other hand, higher vacancy share would mean higher monopsony power and more marked-down wages. The second effect is secondary and is weaker.

The result above holds for other classical labor market power models, if competition

[5] This latter assumption is imposed for the sake of clarity and altering it would not change the results, as shown in the Appendix B.1.

between firms in the market were formulated as in CES, Kimball, nested Kimball and others. In all these cases, γ would be defined as the inverse substitution between labor markets and would be given by: $\gamma_{ijt} = \frac{\partial w_{ijt}}{\partial n_{ijt}} \frac{n_{ijt}}{w_{ijt}}$.

Proposition 1 (**classical labor market power**). *Under models of classical labor market power, there is a positive covariance between equilibrium vacancy shares and wages.*

Proof. See Appendix B.1. □

2.2 Model of Search Labor Market Power

In this subsection we extend the classical labor market power model above with a framework similar to the directed search of Moen (1997). In particular, we assume that consumers are able to observe the wages and vacancies available in each firm-specific labor market and choose which markets to search. Compared to the model of classical labor market power above, consumers will not just need to consider offered wages, but also how many vacancies are available – how likely are they to get the job. Compared to Moen (1997), we simplify the model by having rehiring every period and add imperfect substitution between labor markets as done in classical labor market power models. We then consider the equilibrium relationship between the share of vacancies and wages in this extended model.

General Environment The economy again consists of a representative household and a unit mass of firms. As before, firms hire workers in firm-specific labor markets and produce a differentiable and imperfectly substitutable good to maximize instantaneous profit. Labor markets are now assumed to be subject to search frictions: firms post vacancies and wages and are uncertain about how many workers they will be able to hire. As in the tradition of macroeconomic models (but not usually in directed search models), labor is assumed to be a flexible input, so that firms need to re-hire every period. This assumption greatly aids in considering the two types of oligopsony power in a single, tractable framework. Workers decide on their consumption bundle, choose search effort, and are similarly uncertain about whether they can find employment.

Two Sources of Labor Market Power When choosing their search effort, workers see local labor markets as substitutable based on γ_{ijt}, where ij denotes firm-specific labor market of firm i within a market j. γ_{ijt} is the same source of labor market power as in the baseline classical oligopsony model introduced above. If workers see a firm's labor market as less substitutable, that firm can take advantage of this and mark down wages (see, for instance, Berger et al. 2022 or Card et al. 2018).

In addition and new relative to the previously introduced baseline model, in the presence of search frictions not all workers who search for a job at firm ij will be hired.

Assuming a Cobb-Douglas matching function with matching productivity A_{ij}, firm ij's total hiring is given by $l_{ijt} = A_{ij}\left(\frac{av_{ijt}}{n'_{ijt}}\right)^{\alpha_{ij}} n'_{ijt}$, where n'_{ijt} is the total search time spent by all workers directed at firm ij and v_{ijt} are the vacancies posted by firm ij, and $A_{ij}\left(\frac{av_{ijt}}{n'_{ijt}}\right)^{\alpha_{ij}}$ is the matching rate.[6] Parameter α_{ij} reflects the second source of labor market power, determining how a firm's hiring probability changes when it posts one additional vacancy. A high search labor market power firm, or one with high α_{ij}, is able to offer welfare to the worker through a higher matching rate, which then allows the firm to pay lower wages as shown below. We label this second source of labor market power as *search* labor market power.[7]

Search labor market power can be microfounded as deriving from hiring intensity, which has been found to be a crucial factor driving employment growth at the establishment level (Davis et al., 2013). Firms can adjust hiring intensity by increasing advertising, screening candidates more efficiently, or leveraging economies of scale. The first two align with our model of a fixed parameter α_{ij} that varies across firms, while the latter suggests α_{ij} increases with vacancies, which would result in similar predictions as in our baseline model. Wolthoff (2018) offers a microfounded model of this mechanism, which leads to an amplification of the hiring response just as predicted by our much simpler framework. Lange and Papageorgiou (2020) additionally shows that the aggregate matching elasticity, α, is procyclical, which is consistent with the hiring of higher-alpha firms being more reactive to demand shocks, which would exactly lead to procyclical average α due to a compositional effect. An alternative explanation of search labor market power can be due to differing workers' tolerance for uncertain job-finding probabilities. A firm that faces a pool of applicants that values higher job-finding rates can offer lower wages by posting more vacancies (Barnichon and Figura, 2015; Abbritti and Consolo, 2024).

Household Problem As in the baseline model, we assume the representative household maximizes a utility function that is separable in labor and consumption, which allows us once again to focus on labor market decisions exclusively. The household chooses how much to search (n_{ijt}), taking as given wages (w_{ijt}), total search to firm ij (n'_{ijt}), aggregate search across all labor markets (N_t), and vacancies posted by firm ij (v_{ijt}) according to

[6] Parameter a is calibrated to the reverse long-run market tightness of an average firm, to make sure that in this case hiring is equal to $A_{ij}n_{ijt}$.

[7] When $\alpha_{ij} = 0$, hiring is only influenced by the optimal search condition of the households and not by the number of vacancies posted – so there will be no reason to post vacancies, unless required. Conversely when $\alpha_{ij} = 1$, hiring is not influenced by the optimal search condition of the households, and there will be no connection between wages and hiring, so optimal wages are equal to zero. In practice, α is between zero and one.

the following maximization problem:

$$W_t M_t N_t = \max_{n_{it}} \int_j \left(\sum_i \left(A_{ij} \frac{av_{ijt}}{n'_{ijt}} \right)^{\alpha_{ij}} n_{ijt} w_{ijt} \right) dj$$
$$\int_j \left(\frac{n_{jt}}{N_t} \right)^{\frac{1+\theta}{\theta}} dj = 1 \qquad \sum_i \left(\frac{n_{ijt}}{n_{ij}} \right)^{\frac{1+\eta}{\eta}} = 1$$

where we replaced labor at any given firm with the hiring function introduced earlier comprising the matching rate $A_{ij} \left(\frac{av_{ijt}}{n'_{ijt}} \right)^{\alpha_{ij}}$, which is taken as given by the consumer who does not internalize the "crowding-out" effect of her search efforts, multiplied by the amount of search she chooses; and M_t is the aggregate matching rate, given by $M_t = \int_j A_{ij} \left(\sum_i \left(\frac{av_{ijt}}{n'_{ijt}} \right)^{\alpha_{ij}} n_{ijt} \right) dj / N_t$

The first-order condition for optimal search in each firm's labor market is:

$$A_{ij} \left(\frac{av_{ijt}}{n_{ijt}} \right)^{\alpha_{ij}} \frac{w_{it}}{W_t} \frac{1}{M_t} = \left(\frac{n_{ijt}}{n_{jt}} \right)^{\frac{1}{\eta}} \left(\frac{n_{jt}}{N_t} \right)^{\frac{1}{\theta}} \tag{4}$$

For the above first-order condition, we have used that $n'_{ijt} = n_{ijt}$ in equilibrium, because our household is representative. This optimal search condition defines the labor supply.

Differently from the baseline model, note that now labor supply adjusts to two margins. First, search depends positively on wages, i.e. higher wages in firm ij motivate workers to supply more labor to that firm, as we saw in the previous section. This is the only margin of adjustment in macroeconomic models without search frictions. Second, due to search frictions, finding a job is uncertain, and thus even if firm ij offers high wages, the worker may not direct her search there if the probability of securing a job is too low. Therefore, potential workers trade off the likelihood of finding a job against offered wages. A firm can attract more workers by posting more vacancies, because this raises the probability that any worker will find a job, which is valued by workers.

Firms' Problem Firms maximize profits by choosing which wage to offer and how many vacancies to post, taking the demand for their goods and the amount of search allocated to their labor market as given. The firm's costs now include linear vacancy-posting costs $c_{ij} v_{it}$.[8]

The profit maximization problem for firm i is:

[8]The cost of vacancies can be thought as the cost of sifting through applicants' CVs and doing interviews, as well as expenses tied with incorporating a worker in the firm more generally like training. This cost should not be interpreted literally, since posting a vacancy should be relatively cheap.

$$
\begin{aligned}
&\max_{v_{ijt},w_{ijt}} p_{ijt}y_{ijt} - w_{ijt}l_{ijt} - c_{ij}v_{ijt} - Q_{ijt} \\
&l_{ijt} = A_{ij}\left(\frac{av_{ijt}}{n_{ijt}}\right)^{\alpha_{ij}} n_{ijt} \\
&A_{ij}\left(\frac{av_{ijt}}{n_{ijt}}\right)^{\alpha_{ij}} \frac{w_{ijt}}{W_t}\frac{1}{M_t} = \left(\frac{n_{ijt}}{n_{jt}}\right)^{\frac{1}{\eta}} \left(\frac{n_{jt}}{N_t}\right)^{\frac{1}{\theta}} \\
&y_{ijt} = \mathcal{Y}_{ijt} \\
&y_{ijt} = f(l_{ijt}, I_{ijt})
\end{aligned}
$$

The first constraint denotes the hiring function introduced earlier; the second constraint denotes the optimal labor supply decision from the household problem; and the third and fourth constraints denote the demand and the production function, respectively. As before, this problem allows for capital to be an additional input into production.[9]

γ_{ijt} is defined exactly as in the baseline model above:

$$
\gamma_{ijt} = \frac{1}{\eta} + \lambda_{ijt}\left(\frac{1}{\theta} - \frac{1}{\eta}\right), \quad \text{where } \lambda_{ijt} = \frac{l_{ijt}w_{ijt}}{\sum_i l_{ijt}w_{ijt}}
$$

However, because of search frictions, hiring, vacancies posted and search efforts are now all distinct and thus while γ_{ijt} is still defined as an inverse substitution rate between labor markets, it will no longer be equal to the inverse elasticity of labor supply.

There are now two important elasticities: the elasticity of labor supply with respect to wages, the only elasticity that mattered in the baseline classical oligopsony model of the previous subsection, and a new elasticity – the elasticity of labor supply with respect to vacancies:

$$
\xi^w_{ijt} = \frac{\partial l_{ijt}}{\partial w_{ijt}}\frac{w_{ijt}}{l_{ijt}} = \frac{1-\alpha_{ij}}{\gamma_{ijt}+\alpha_{ij}}, \qquad \xi^v_{ijt} = \frac{\partial l_{ijt}}{\partial v_{ijt}}\frac{v_{ijt}}{l_{ijt}} = \alpha_{ij}\frac{\gamma_{ijt}+1}{\alpha_{ij}+\gamma_{ijt}}
$$

The first-order conditions of the problem above are:

$$
v_{ijt} = \frac{\xi^v_{ijt}}{\xi^w_{ijt}+1}\frac{m_{ijt}}{c_{ij}}l_{ijt}, \qquad w_{ijt} = \frac{\xi^w_{ijt}}{\xi^w_{ijt}+1}m_{ijt}
$$

where $m_{ijt} = (p_{ijt}y_{ijt})'_l$ is the marginal revenue of labor, which again subsumes many important firm-level characteristics like productivity and markups.

Solving for the elasticities using the household optimal search equation, the first order

[9] Again that would be done through the parameter vector I_{ijt} and investment costs are included in Q_{ijt}. The only assumption made is that labor and capital spending are separable in the cost function, so that Q_{ijt} does not depend on labor.

conditions above simplify to:

$$v_{ijt} = \alpha_{ij}\frac{m_{ijt}}{c_{ij}}l_{ijt}\ , \qquad w_{ijt} = \frac{1-\alpha_{ij}}{\gamma_{ijt}+1}m_{ijt} \tag{5}$$

The latter is the same as equation (9) of Manning (2021) using a different framework based on directed search and the classical labor market power model based on preferences for amenities of Card et al. (2018). An equation similar to the above is present in the contemporaneous work of Berger et al. (2024), with m_{ij} in their case being the aggregate surplus value for hires of firm ij. Importantly, Berger et al. (2024) does not consider the impact of vacancies on the matching rate for all searching workers – thereby abstracting from the margin of adjustment that we emphasize in this study.[10]

The markdown, or the extent to which firms pay wages below the marginal revenue of labor, is given by a modified Lerner formula (compare to equation 3):

$$\nu_{ijt} = \frac{\xi^w_{ijt}}{\xi^w_{ijt}+1} = \frac{1-\alpha_{ij}}{\gamma_{ijt}+1} \tag{6}$$

The two sources of monopsony introduced earlier are visible in the formula for the markdown. The first derives from *classical* labor market power and is more commonly discussed in the literature (e.g., Berger et al. 2022). It is represented by γ_{ijt}, which determines the inverse of the elasticity of substitution between labor markets for the worker. Under classical labor market power, all else equal, wages depend negatively on γ_{ijt}, meaning that firms are able to pay lower wages if switching between labor markets is difficult. The second source of labor market power is due to the presence of search frictions and is reflected in the parameter α_{ij}. Under search oligopsony, wages depend negatively on the rate through which a firm converts vacancies to hiring as governed by α_{ij}. Firms are able to pay lower wages because they can offer an alternative source of value to workers: by posting vacancies, they can reduce the likelihood of unemployment for workers searching in their labor market. Markdowns arise from a combination of these two forces. Note that the markdown formula collapses to that in the baseline model of the previous subsection if $\alpha = 0$, $c = 0$, and $v_{ijt} = l_{ijt}$.

To describe the equilibrium, we consider heterogeneity across firms in terms of marginal revenue from labor m_{ijt}, labor supply parameters η and θ embedded in γ_{ijt}, matching elasticity α_{ij}, matching productivity A_{ij} and vacancy-posting costs c_{ij}; while keeping other firms' variables and economy-wide aggregates fixed. We look at the relationship between

[10]The formula from Berger et al. (2024) can be stylized as follows: $c'(v) = A\left(\frac{v}{n}\right)^{\alpha-1}\mathbb{S}$ with $\mathbb{S}$ being the average surplus of hiring, an intuitive counterpart of m, and $c'(v)$ the derivative of the cost of posting vacancies with respect to vacancies, in our case equal to c_{ij}. The corresponding formula in our model can be written as follows: $c = \alpha A\left(\frac{v}{n}\right)^{\alpha-1}m$

equilibrium vacancies and wages that results from this heterogeneity.

In this setting there is a distinction between search, hiring and vacancies posted, so the relationship between equilibrium vacancies and wages is no longer determined exclusively by the slope of the labor supply.

As before, consider a log-linearization of the labor supply (equation 4) and equilibrium conditions (equations 5) around the average firm in an average market, such that $n_{ijt} = n_{jt} = N_t$, equilibrium market tightness is scaled so that $\frac{av_{ij}}{n_{ij}} = 1$,[11] and keeping other firms n_{kjt} and aggregate variables fixed. As before, this latter assumption is imposed for the sake of clarity and altering it would not change the results, as shown in the Appendix B.2.

We consider heterogeneity across firms along several dimensions: (i) marginal revenue from labor m_{ijt}, (ii) matching elasticity parameter α_{ij}, (iii) matching productivity A_{ij}, and (iv) vacancy-posting cost c_{ij}. We look at the relationship between equilibrium vacancies and wages that results from this heterogeneity. The following denotes the resulting expression for wages and the share of vacancies, where tildes denote departures from the average firm in the average market:

$$\tilde{w}_{ij} = \gamma \tilde{n}_{ij} - \underbrace{(\tilde{A}_{ij} + \alpha(\tilde{v}_{ij} - \tilde{n}_{ij}))}_{\text{change in matching rate}}$$

$$\tilde{s}_{ij} = (1-s)\tilde{v}_{ij} \qquad \tilde{l} = \tilde{A}_{ij} + \alpha_{ij}\tilde{v}_{ij} + (1-\alpha)\tilde{n}_{ij}$$

The decision on vacancies does not depend on the search directed to other firms, so the covariance between the share of vacancies and wages is:

$$\text{cov}(\tilde{w}_{ij}, \tilde{s}_{ij}) = (1-s)\frac{1}{\xi^w}\text{cov}(\tilde{l}_{ij}, \tilde{v}_{ij}) - (1-s)\frac{1}{\nu}\text{var}(\tilde{A}_{ij}, \tilde{v}_{ij}) - (1-s)\alpha\frac{1}{\nu}\text{var}(\tilde{v}_{ij})$$

where $\text{cov}\left(\tilde{l}_{ij}, \tilde{v}_{ij}\right), \text{cov}\left(\tilde{A}_{ij}, \tilde{v}_{ij}\right)$ are positive. Proofs and complete formulas are available in the Appendix B.2. The formula above collapses to the same covariance in the baseline model without search if $\alpha = 0$, $A = 1$ and $v_{ij} = l_{ij} = n_{ij}$, in which case $\text{cov}(l_{ij}, v_{ij}) = \text{var}(\tilde{n}_{ij})$ and $1/\xi^w = \gamma$

Like in the baseline case, if search labor market power is limited (α_{ij} is close to zero), and there is not much variation in matching productivity A_{ij} across firms, the positive covariance between vacancies and wages will persist since the covariance between hiring and vacancies is positive and would outweigh the small negative terms at the end of the expression. While in the above we allow for heterogeneity in the marginal revenue of labor m_{ij}, vacancy posting costs c_{ij}, matching intensity A_{ij}, and α_{ij}, only variation in α_{ij} and in

[11]Note from the labor supply equation that only $A\left(\frac{av}{n}\right)^{\alpha}\frac{w}{WM}$ matters for search effort n_{ijt}, so this serves to scale the matching productivity so that for the average firm $l = n$.

matching intensities A_{ij} can generate a negative relationship between wages and vacancy shares. Neither are present in the baseline model without search labor market power. At this stage, one cannot be sure whether the negative covariance is caused by α_{ij}, or A_{ij}, so studying this covariance would only serve to reject the model without search market power but would not discern the relative importance of α_{ij} or A_{ij}.

Thus if α_{ij} rises, wages would decrease and vacancies would increase if this effect is sufficiently strong, so there can be a negative covariance between vacancies and wages. Proposition 2 states this result more formally.

Proposition 2 (**search monopsony**). *Firms with greater search monopsony (larger α_{ij}) pay lower wages, but post **more** vacancies in equilibrium. There will be a negative covariance between vacancies and wages if search effects are sufficiently strong.*

Proof. See Appendix B.2. □

As before, the result of this proposition holds for multiple models of classical monopsony with different substitution between labor markets, e.g. CES, Kimball, among others. In contemporaneous work that combines search and classical monopsony, the model of Berger et al. (2024) would also generate a negative covariance between wages and vacancies, however that paper assumes this effect away by abstracting from the effect of vacancy posting on matching.

It follows from Proposition 2 that both sources of labor market power allow firms to mark down wages, as also evident from equation (6), but have opposite effects on vacancies. Firms with classical monopsony internalize that workers cannot easily switch away from their labor market and take advantage of that by offering lower wages and lower vacancies, which are both costly for the firm and beneficial for the worker. Firms with search monopsony offer more vacancies to the extent that is cheaper than raising wages.

Propositions 1 and 2 suggest the following preliminary test of the relative prevalence of the two sources of labor market power:

Test 1: In equilibrium, classical labor market power implies a positive relationship between vacancies and wages, while search labor market power implies a negative relationship. This is tested in Section 4.1.

Test 1 does not distinguish the role of α_{ij} and A_{ij}. In the next section, we are able to discern the relative importance of α by considering exogenous demand shocks.

2.3 Monetary Policy Response

Empirically, it is difficult to disentangle the sources of labor market power in equilibrium. In this subsection, we show how the relative importance of classical and search labor market power can be assessed using the response of vacancy postings and wages to exogenous demand shocks. To do so, we use exogenous monetary policy shocks.

Monetary policy shocks are widely regarded as effective instruments for identifying exogenous demand shocks. Central banks implement changes in monetary policy, such as adjustments in interest rates or quantitative easing, with the explicit goal of influencing aggregate demand. Using the unanticipated part of the monetary policy decision allows us to treat them as exogenous variations of aggregate demand. By analyzing vacancies and wages in response to these shocks, we can infer the effects of demand-side disturbances while minimizing concerns about reverse causality or omitted variable biases.

In this environment, monetary policy does not affect hiring directly. Instead, monetary policy affects the hiring problem indirectly through the marginal revenue of labor, or m_{ijt}. This follows from assuming a static hiring problem and means that monetary policy only affects household consumption (the demand for firm's goods) and firms' investment.

In this case, we take the equilibrium as given and allow for all the parameters to vary between firms. We consider log-deviations from the equilibrium in equation 5 and look at a small change in m_{ijt} over time denoted with a hat. Given the response of wages $\hat{w}_{ijt}$, the contemporaneous responses of vacancies is given by:

$$\begin{aligned}\hat{v}_{ijt} =& \frac{1}{1-\alpha_{ij}}\hat{m}_{ijt} + \eta\left(\frac{\alpha_{ij}}{1-\alpha_{ij}}\hat{m}_{ijt} + \hat{w}_{ijt} - \sum_i \lambda_{ij}\left(\frac{\alpha_{ij}}{1-\alpha_{ij}}\hat{m}_{ijt} + \hat{w}_{ijt}\right)\right) + \\ & \theta\left(\sum_i \lambda_{ij}\left(\frac{\alpha_{ij}}{1-\alpha_{ij}}\hat{m}_{ijt} + \hat{w}_{ijt}\right) - \int_j \lambda_j \sum_i \lambda_{ij}\left(\frac{\alpha_{ij}}{1-\alpha_{ij}}\hat{m}_{ijt} + \hat{w}_{ijt}\right) dj\right) + \hat{N}_t\end{aligned} \tag{7}$$

where λ_{ij} is the equilibrium payroll share, $\lambda_{ij} = \frac{w_{ij}l_{ij}}{w_j l_j}$, and $\lambda_j = \frac{w_j l_j}{WL}$.

Responses under the baseline model without search frictions can be obtained by setting α_{ij} to zero and by dropping the first term because under that model $v/n \equiv 0$. Under such a model, the response of vacancies is $\hat{v} = \hat{N}_t$ and vacancies respond equally across all firms.

The absence of heterogeneity in the response of vacancies persists, unless there is heterogeneity in α_{ij} or heterogeneity in the response of wages $\hat{w}_{ij}$. This holds even if we allow for heterogeneity in parameters η and θ, or consider a Kimball-style substitution. Intuitively, this result stems from the assumption of homogeneity of degree one in the substitution between labor markets, that is, from the assumption that search allocated by households only takes into account relative (real) wages. This result is stated formally in the Proposition 3.

Proposition 3 (monetary policy response). *In a model without search labor market power and following a monetary policy shock, if firms with varying degrees of classical labor market power do not change their wages differentially, vacancies of these firms also do not change differentially. However, if firms have varying degrees of search labor market power and do not change their wages differentially, vacancies of firms with greater search labor market power react more.*

Proof. See Appendix B.3. □

Crucially for Proposition 3 is that wages do not respond differentially to either classical or search labor market power. Under this condition, two firms with the same level of search labor market power and different levels of classical labor market power would also not post vacancies differentially because the elasticity of labor supply to vacancies is the same for both. On the contrary, if there are two firms that differ in search labor market power, they would adjust their vacancies differentially, because the firm with greater search labor market power can gain more by posting one additional vacancy.

We would also note that if wages respond homogeneously across firms with varying degrees of either classical or search monopsony power, that rules out that differences in the marginal revenue of labor would be the driving factor behind the response of vacancies, because any such differences would be reflected in the response of wages.

Appendix B.4 shows that Proposition 3 is robust to non-constant returns to scale or imperfect competition in the product market.

Importantly, Appendix B.5 shows that Proposition 3 is robust to assuming wages are rigid as long as wages respond homogeneously to varying degrees of both classical and search monopsony power.[12] This is relevant since even though there is some evidence of relative flexibility of wages of new hires.[13]

To test Proposition 3, we formulate a test based on responses to monetary policy shocks using data on *vacancies and wages*, because we do not observe directly the degree of either search or classical labor market power in the data and estimates of either parameter are scant in the literature.

To formulate a testable hypothesis, we impose the assumption of homogeneous wage responses and consider deviations of the response of vacancies for any given firm relative

[12]Note that no additional assumptions on the wage setting mechanism are needed, so the proposition holds under time-dependent (*a la* Calvo) and state-dependent (menu costs) frameworks.

[13]See Haefke and Sonntag (2013) and Martins et al. (2012). More recent papers, such as Grigsby et al. (2021) and Hazell and Taska (2020), have found evidence of substantial wage rigidities even for new hires. Grigsby et al. (2021) shows that time-dependence (similar to Calvo pricing) is a good approximation of how wages adjust outside recessions.

to the average firm in the market to a shock to $\hat{m}$. Denote the deviations from the average firm, as before, with tildes, for instance, $\tilde{\alpha}_{ij}$ denotes the log deviation of the parameter α of firm ij from the average firm. Since all firms are linearized around the average, equation (7) simplifies greatly and becomes:

$$\hat{v}_{ijt} = \frac{\alpha}{1-\alpha}\tilde{\alpha}_{ij}\hat{m}_t + \hat{N}_t$$

Now, consider the coefficient from running a simple regression of vacancies on the monetary policy shock and its interaction with the deviation of equilibrium vacancy shares $\tilde{s}_{ij}$ and equilibrium wages $\tilde{w}_{ij}$ from those of the average firm. Assuming that monetary policy shocks are independent of equilibrium vacancies and wages and recalling that deviations from the average firm have zero expected value by construction, the regression coefficients would be:

$$\beta^{vv} = \frac{\text{cov}(\hat{v}_{ijt}, \tilde{s}_{ij}\hat{m}_t)}{\text{var}(\tilde{s}_{ij}\hat{m}_t)} = \frac{\alpha}{1-\alpha}\frac{1-s}{1-\alpha}\frac{\text{var}(\tilde{\alpha}_{ij}\hat{m}_t)}{\text{var}(\tilde{s}_{ij}\hat{m}_t)} > 0$$

$$\beta^{vw} = \frac{\text{cov}(\hat{v}_{ijt}, \tilde{w}_{ij}\hat{m}_t)}{\text{var}(\tilde{w}_{ij}\hat{m}_t)} = -\left(\frac{\alpha}{1-\alpha}\right)^2\frac{\text{var}(\tilde{\alpha}_{ij}\hat{m}_t)}{\text{var}(\tilde{w}_{ij}\hat{m}_t)} < 0$$

Hence, if α varies across firms, the coefficient on the interaction between monetary policy shocks and equilibrium vacancies will be *positive*, while that on the interaction of monetary policy shocks and equilibrium wages will be *negative*.[14] With these effects in mind, we are now ready to formulate our second test:

Test 2: If wages of all firms respond homogeneously to monetary policy and vacancies respond more in firms with higher equilibrium vacancies and lower equilibrium wages, the model where firms vary only in their extent of classical monopsony can be rejected. This implies that search monopsony drives the response of wages and vacancies to monetary policy. This is tested in Section 4.2.

3 Data

3.1 Lightcast data

Lightcast data, formerly Burning Glass Technologies, tracks online vacancy postings from over 45,000 online job boards, carefully removes duplicates, and cleans the data. The resulting dataset covers the near universe ($\approx$ 70%) of all U.S. vacancy postings and comprises $\approx$ 250 million job vacancy postings for the years of 2007 and 2010-2019.

[14] Note additionally, that we use equilibrium vacancies and wages instead of directly observed α, which biases the coefficients β^{vv} and β^{vw} towards zero. This bias is greater if variation in α explains a small share of the variation in equilibrium vacancies and wages.

Lightcast has an extensive coverage across differently sized firms. Unlike survey data, it is collected directly from firms' postings and therefore is a more accurate representation of the vacancies in the economy. Concretely, it is free from the limitations of datasets that only cover firms of a certain size or firms that satisfy certain criteria, such as being publicly traded like Compustat.

Lightcast can be used as an establishment level data due to extensive geographical breakdown of posted vacancies. Each vacancy contains a FIPS county code, which allows it to be mapped onto any other geographical unities, such as states, MSA or, in our case Commuting Zones.

Lightcast data also offers other significant details on the type of vacancy. NAICS industries, and ONET occupation breakdowns are available. A large proportion of vacancies also lists job requirements, such as education or software skills. Education is reported for approximately half of vacancies. When education is missing, we impute it based on the data for the existing vacancies using the finest occupational breakdown. Effectively we assign the same education requirement within the same occupation. This procedure eliminates most of the missing values.

Lightcast vacancy data has some shortcomings due to the way it is collected, especially in earlier years. The main concern is that online vacancy postings are not representative of all the postings in the economy with an over-representation of certain industries, such as IT or Education. However, robustness checks, for instance in Hershbein and Kahn (2018) indicate that, despite these shortcomings, the resulting vacancy data tracks aggregate and industry trends closely.

Lightcast wage data is significantly less extensive than vacancy data, with only 17% of the vacancies reporting wages. Hazell et al. (2021) find that this limitation does not preclude the data from being representative, with median posted wages from Lightcast closely tracking the corresponding statistics in Occupational Employment Statistics (OES). Some postings list a wage range — in these instances, we take the midpoint of that range. For most of our analysis, we collapse vacancy-level data into a panel of firm-, commuting zone- and quarter-level, or effectively an establishment-level panel. Wage data on annual compensation and does not include bonuses and other benefits beyond the basic wage.

Definition of local labor market. Our analysis mainly compares responses of establishments within the same labor market, so it is crucial to both define a local labor market and come up with equilibrium vacancy and wage measures that are consistent. We define a local labor market as a commuting zone, because it is a closer representation of the options available to a worker searching for job in a particular labor market.

Measure of equilibrium vacancies and wages. As discussed below, we use cumulative vacancy shares as an equilibrium measure of vacancies and cumulative wage residual, cleaned from occupational effects, as an equilibrium measure of wages.

The measure of the equilibrium level of vacancies is the cumulative share of vacancies posted by a single firm in a local labor market out of the total vacancies posted in that labor market:

$$\text{Equilibrium Vacancy Share}_{i,c,t} = \frac{\sum_{\tau<t} v_{i,c,\tau}}{\sum_{\tau<t}\sum_i v_{i,c,\tau}} \tag{8}$$

where vacancies in commuting zone c, for firm i at time τ are denoted by $v_{c,i,\tau}$.

To measure the equilibrium level of wages we need to account for the significant heterogeneity across occupations and geography. First, we residualize wages by running a fixed-effect regression and canceling out any occupation and commuting zone related factors; and aggregate the results to firm-CZ-time level. We then compute a vacancy-weighted cumulative wage measure using the formula below and normalize the result:

$$\text{Equilibrium Wage Measure}_{i,c,t} = \frac{\sum_{\tau<t} v_{i,c,\tau} \times \text{residual wages}_{i,c,\tau}}{\sum_{\tau<t} v_{i,c,\tau}} \tag{9}$$

where residualized wages in commuting zone c, for firm i at time τ are denoted by $\text{residual wages}_{i,c,\tau}$ and vacancies of firm i in commuting zone c at time τ is denoted by $v_{i,c,\tau}$

Dependent variables. As a dependent variable, we use log-vacancies and deviations of wages from local average to more closely track the difference of particular establishments from its labor market competitors in the local labor market, the latter distinction, however, does not play a role once we include commuting zone - time fixed effects.[15]

Our unit of analysis is the firm-commuting zone-time level. We have in total over 15 million firm-commuting zone-quarter observations with a total of over 380,000 firms and just over 700 commuting zones. The average commuting zone has postings from 22,000 firms, although this is unevenly distributed. The average firm posts in 170 commuting zones (see Table 1 for further details).

3.2 Monetary Policy shocks

The baseline measure of monetary policy shocks we use is that developed in Jarociński and Karadi (2020) — JK 2020 henceforth. They focus on interest rate surprises in three-

[15] Average regional wages are subtracted to abstract from compositional effects, for example, after a contractionary monetary shock, postings in high wage regions (low labor market power) fall by less, biasing the response of wages up. Subtracting average regional wages allows partialing out commuting zone time characteristics.

month fed funds futures, which exchange a constant interest for the average federal funds rate over the course of the third calendar month in the contract. As regular FOMC meetings are 6 weeks apart from each other, the three-month future reflects the shift in the expected federal funds rate after the following policy meeting. These shocks do not capture surprises to the balance sheet, implicitly assuming that such changes are orthogonal to surprises to the policy rate (and that balance sheet measures would not affect the 3-month futures).

We prefer JK 2020 shocks because they separate pure monetary policy shocks from signaling shocks related to the state of the economy — so called "Fed information" (Romer and Romer, 2000; Nakamura and Steinsson, 2018) or "Fed reacting to information" (Bauer and Swanson, 2023) shocks. The first reflects the fact that economic agents take Fed actions as a signal about the state of the economy and adjust their expectations accordingly. For instance, a surprising monetary loosening can be taken as a sign that the economy is performing poorly and as a result economic agents might, for instance, reduce investment. The "Fed reacting to information" interpretation of the same loosening is that the counter-intuitive response to monetary policy results from the failure to control for recent bad news on the economy known to both the Fed and agents. The effect of Fed information shocks, therefore, goes in the opposite direction to that of monetary policy, and mixing the two together can significantly bias the results and confound channels. As a baseline, we are only interested in the effect of the monetary policy shock and we use the information shock as a control.[16]

Appendix Table A7 and Table A8 show that our main results are robust to using other measures of monetary policy shocks, including those of Nakamura and Steinsson (2018), Jarocinski (2021), and Bu et al. (2021).[17]

[16] Appendix Figure A1 presents the time-series of JK 2020 shocks, both monetary policy and central bank information. Monetary policy shocks exhibit both significant tightening and loosening. Our vacancy dataset includes years 2007 and 2010-2019. Over this period, the largest tightening and loosening shocks happened in 2007. The global financial crisis-related loosening cycle started in August 2007 and in the run up to it uncertainty over the state of the economy and thus over monetary policy was elevated.

[17] Nakamura and Steinsson (2018) use principle component analysis to combine in one shock surprises across the yield curve from one-month to two-year. Jarocinski (2021) estimates four different shocks, including the standard monetary policy shock and three orthogonal shocks that do not affect near-term fed funds futures. The other shocks include: (i) an Odyssean forward guidance shock (a commitment to a future course of policy rates); (ii) a shock to longer-term treasury yields mostly affected by asset purchase announcements; and (iii) a Delphic forward guidance shock (Campbell et al., 2012), which captures the stance of future monetary policy in the sense of a prediction of the appropriate stance of policy, rather than its commitment. Bu et al. (2021) include unconventional policy constructed through a Fama-MacBeth two-step procedure to extract monetary policy shocks from the common component of outcome variables. They conclude that their measure does not contain a significant central bank information effect.

4 Empirical Results

The model in section 2 provides two tests: (1) on equilibrium vacancies and wages; and (2) on responses of vacancies and wages to monetary policy shocks. In this section, we use data on vacancies, wages, and monetary policy shocks (as exogenous demand-shifters) to test which is the dominant source of labor market power across firms — classical monopsony or search monopsony. We find that: (1) in the cross-section, firms that post more vacancies in a regional labor market offer lower wages even after controlling for many observed and unobserved characteristics; and (2) firms that account for more vacancies and post lower wages in a regional labor market raise vacancies by more following monetary policy loosening, without having to increase wages. Both findings are inconsistent with the predictions from a model with classical labor market power only and suggest that search monopsony may be the dominant source of variation in labor market power in the data.

4.1 Test 1: Equilibrium Vacancies and Wages

Test 1: *In equilibrium, classical monopsony implies a positive relationship between vacancies and wages, while search monopsony implies a negative relationship.*

Classical monopsonists curtail vacancies and employment because they internalize that all workers are paid more when the firm offers higher wages to attract new hires. This creates higher markdowns on their wages, implying a positive correlation between vacancies and wages (Proposition 1). In contrast, search monopsonists post more vacancies allowing them to reduce wages, as workers value the higher employment probability, implying a negative correlation between vacancies and wages (Proposition 2). This fundamental dichotomy suggests that studying the cross-sectional correlation of wages and vacancies in the data can provide preliminary evidence on the importance of search market power relative to classical monopsony.

Indeed, we find that a firm's share of vacancies in a given commuting zone is correlated with *lower* posted wages, suggesting an important presence of search monopsony. Figure 1 plots firms' commuting zone-level average posted wage on the y-axis against the vacancy share of that same firm in the same commuting zone on the x-axis. To account for the right-skewed nature of vacancy shares, the x-axis uses a log-scale. The left panel plots the relationship for non-college vacancies and the right panel for college vacancies. A large portion of the distribution of vacancy shares offers similar posted wages. In particular, firms with a vacancy share between 0 and 0.00005 (0.005%) post average wages between US$50,000 and US$48,000 for non-college vacancies. These firms have extremely low labor market power, and within that group differences are not meaningful. A firm that accounts for a vacancy share of 0.00005 posts 1 out of 20,000 vacancies. However, once a firm starts to control more than 0.005 % of the market their posted wage declines strongly. For instance, a firm with a vacancy share of 0.1%, posting 1 in every 1000 vacancies, posts an average wage of less than US$45,000 for non-college vacancies. The same pattern is visible for college vacancies, with posted wages declining more linearly than for non-college vacancies, but with a large drop in posted wages at vacancy shares of about 0.005%. As expected, the overall level of wages posted is significantly higher at around US$76,000 for firms with a vacancy share of < 0.00005) and around US$70,000 for firms with a vacancy share of > 0.001. The negative relationship between vacancies and wages could be spurious or suffer from compositional biases. For instance, if firms with high vacancy shares hire less-skilled workers, lower wages would not be directly due to their higher vacancy share. The split between college vs. non-college vacancies partly addresses this compositional issue, making only a within college/non-college comparison, but cannot fully dismiss compositional issues, as even within each level of educational attainment, skill requirements and productivity can differ significantly.

We test the importance of search monopsony more formally by estimating vacancy-level regressions that control for a large set of vacancy characteristics, such as commuting zone - time and ONET occupation - time fixed effects. We find that even after controlling for observed and unobserved vacancy, firm, and region characteristics, firms with higher vacancy shares post lower wages on their vacancies (Table 2). We interpret this result as evidence against the model with only classical market power and in favor of the model with search labor market power. We note that any confounding factors of firm-level productivity that may be left in these regressions would bias our estimates towards finding a positive correlation between vacancies and wages.

Confirming a negative relationship between vacancies and wages, we can use the result of Proposition 2, to perform a back of the envelope calculation and obtain the lower bound for the average α across firms. Using the simplified formula and assuming that there is no variation in matching efficiency $\tilde{A}_{ij}$, the covariance between vacancies and wages would be given by: $\text{cov}(w, v) = \frac{1}{\xi^w}\text{cov}(\tilde{v}_{ij}, \tilde{l}_{ij}) - \alpha\frac{1}{\nu}\text{var}(\tilde{v})$. Using the variance of vacancies and the covariance between vacancies and employment from the merged Lightcast-Compustat dataset, and using the range of values for ξ^w from Berger et al. (2022), we obtain: $\alpha > 0.18$. This would mean that α is responsible for at least 54% of the average markdown.

4.2 Test 2: Response to Monetary Policy Shocks

Model Predictions and Empirical Approach

Test 2: *If wages of all firms respond homogeneously to monetary policy and vacancies respond more in firms with higher equilibrium vacancies and lower equilibrium wages, the* *model where firms vary only in their extent of classical monopsony can be rejected. This implies that search monopsony drives the response of wages and vacancies to monetary policy.*

To implement Test 2, we run the following two specifications:

$$\log vacancies_{i,c,t} = \beta^{vv} \text{ MP easing}_t \times \text{equilibrium vacancies}_{i,c,t-1} + \Theta^{vv}\text{controls} + \epsilon^{vv}_{i,c,t} \tag{10}$$

$$\log vacancies_{i,c,t} = \beta^{vw} \text{ MP easing}_t \times \text{equilibrium wages}_{i,c,t-1} + \Theta^{vw}\text{controls} + \epsilon^{vw}_{i,c,t} \tag{11}$$

where i is a firm, in a local labor market c at time t.[18]

Recall from Proposition 1, classical monosponists offer lower equilibrium vacancies and lower wages. Instead, Proposition 2 suggests search monopsonists offer *higher* equilibrium vacancies and lower wages. Putting these together with Proposition 3 implies that ignoring differences in search monopsony across firms, it should be that we find $\beta^{vv} = \beta^{vw} = 0$, as firms with classical monopsony power should not react differentially to monetary policy shocks. If instead, we allow for heterogeneity in search, proposition 3 states firms with higher equilibrium vacancies and *lower* equilibrium wages (more labor market power) post more vacancies in response to a monetary policy shock. Hence, we should find that $\beta^{vv} > 0$, but $\beta^{vw} < 0$. This is the essence of Test 2.

In addition, Test 2 requires that wages do not respond differentially to either classical or search market power. If that were not the case, estimates of β^{vv} and β^{vw} in equations (10) and (11) could reflect non-labor market characteristics that vary across firms, crucially firm productivity rather than labor market power. To assure that the conditions for Test 2 hold, we run the following two additional regressions:

$$\log wages_{i,c,t} = \beta^{wv} \text{ MP easing}_t \times \text{equilibrium vacancies}_{i,c,t-1} + \Theta^{wv}\text{controls} + \epsilon^{wv}_{i,c,t} \tag{12}$$

$$\log wages_{i,c,t} = \beta^{ww} \text{ MP easing}_t \times \text{equilibrium wages}_{i,c,t-1} + \Theta^{ww}\text{controls} + \epsilon^{ww}_{i,c,t} \tag{13}$$

where again i is a firm, in a local labor market c at time t.[19]

Test 2 requires the interaction term between the monetary policy shock and equilibrium characteristics to be zero, $\beta^{wv} = \beta^{ww} = 0$.

[18]**controls** include: (i) the Federal Reserve information shock and its interactions with equilibrium vacancies or wages, respectively, (ii) equilibrium vacancies or wages depending on whether equation (10) or (11) are estimated, (iii) firm fixed effects that absorb any time-invariant firm variation, and (iv) commuting zone–time effects that absorb any time-varying regional shocks, such as local demand shocks.

[19]**controls** include: (i) the Federal Reserve information shock and its interactions with equilibrium vacancies and wages, (ii) equilibrium vacancies or wages depending on whether equation (12) or (13) are estimated, (iii) firm fixed effects that absorb any time-invariant firm variation, and (iv) commuting zone–time effects that absorb any time-varying regional shocks, such as local demand shocks.

Main Results

Table 3 shows estimates for $\beta^{vv}, \beta^{vw}, \beta^{wv}$, and β^{ww} as defined in equations (10)-(13).

In a model with search, labor market power amplifies the effect of monetary policy on vacancies. Instead, in a model with classical labor market power, the response of vacancies would not vary.

The first column estimates a positive and significant β^{vv}. In response to an accommodative (contractionary) monetary policy shock, firms that account for a larger share of regional vacancies increase (decrease) their vacancy postings by more than their counterparts. This result is consistent with firms with higher search labor market power responding more to monetary policy shocks. Firms with higher classical labor market power would have lower equilibrium vacancy shares and their differential vacancy response would be muted.

Column (2) of Table 3 estimates β^{vw}, or whether firms that pay relatively lower wages compared to their counterparts react differently to monetary policy in terms of their vacancy postings. Under both classical and search labor market power, lower wages are associated with more labor market power. β^{vw} is estimated to be negative and significant, meaning that firms that pay *lower* wages also increase vacancy postings more in response to monetary policy easing.

These estimates strongly suggest that heterogeneity in search labor market power is a key driver of vacancies and not classical labor market power, which would predict a null result in both columns (1) and (2).[20]

Finally, columns (3) and (4) of Table 3 present estimates for equations (12) and (13). We cannot reject that β^{wv}, and β^{ww} are both zero, i.e. we cannot find a differential wage response to monetary policy of firms with either higher share of vacancies or higher wages. These support our conclusion that columns (1) and (2) are driven by heterogeneity in search monopsony power rather than other potential non-labor confounding factors.

Alternative assumptions

Table 4 re-estimates column (1) of Table 3 under alternative assumptions. In column (1) we include firm fixed effects to control for unobserved and observed time-invariant heterogeneity at the firm-level, for instance, the average number of vacancies a firm posted during our sample period. It also includes commuting zone fixed effects to control for time-invariant heterogeneity at the commuting zone level. Note that the exclusion of any type of time fixed effects allows us to identify directly the effect of monetary policy shocks, which is collinear in a specification once time fixed effects are included. As

[20]Note that Lightcast data for posted wages is much less comprehensive than data on vacancies and that explains the drop in observations between column 1 and 2.

expected, monetary policy easings are associated with more vacancy postings. Quantitatively, a 10 bp monetary policy easing is associated 7% higher vacancy postings. This effect is amplified when firms have a larger share of vacancies, as already shown in column (1) of Table 3 under a different set of fixed effects. Column (2) replaces commuting zone fixed effects with time fixed effects and column (3) has firm, commuting zone, and time fixed effects. Column (4) shows in comparison our baseline specification and column (5) introduces firm-time fixed effects, which allows comparing vacancy posting within firms across commuting zones. The interaction coefficient is always positive and statistically significant, indicating that firms with higher vacancy shares are more responsive in terms of their vacancy postings than their counterparts. Quantitatively, the interaction coefficient is relatively stable varying between 5.4 and 7.9, implying that firms with a 1 percentage point higher share of regional vacancies have a between 0.54 and 0.79 pp stronger response to a 10 bp monetary policy shock. For instance, a 10 bp accommodative monetary policy shock would increase vacancy postings by 7% for a firm without any vacancy postings, while for a firm that has 1% vacancy share (or the 84th percentile in the vacancy-weighted distribution of vacancy shares) the response is between 7.53% and 7.79%.

Table 5 re-estimates column (2) of Table 3 under alternative assumptions. The first column estimates the specifications only with firm and commuting zone fixed effect. The effect of monetary policy shocks on vacancy postings is amplified for firms that pay low wages, consistent with column (2) of Table 3. Column (2) replaces commuting zone fixed effects with time fixed effects and column (3) has firm, commuting zone, and time fixed effects. Column (4) shows in comparison our baseline specification and column (5) introduces firm-time fixed effects, which allows comparing vacancy posting within firms across commuting zones. The interaction coefficient is always negative and statistically significant, except for column (5), indicating that firms with lower wages are more responsive in terms of their vacancy postings than their counterparts. One potential explanation for why the coefficient becomes insignificant in column (5) is that some firms set wages nationally, as pointed by Hazell et al. (2021). In this case, firm-time fixed effects would absorb any wage variation, rendering the estimated coefficient insignificant and very close to zero. This mechanical feature, however, does not influence the results of the regressions with the vacancy share interaction, as vacancies vary across a firm's locations. National wage-setting would only weaken the mechanism we highlight, but not undo it. Except for the last column, quantitatively, the interaction coefficient is relatively stable varying between -0.058 and -0.067, implying that firms with one standard deviation lower wages have between 0.58 and 0.67 pp stronger response to a 10 bp monetary policy shock. For instance, a 10 bp accommodative monetary policy shock would increase vacancy postings by 8.7% for the average firm, while for a firm that has one standard deviation lower wages the response is between 9.28% and 9.37%.

Table 6 re-estimates column (3) of Table 3 under alternative assumptions. The first column estimates the specifications only with firm and commuting zone fixed effects. This allows us to identify directly the effect of monetary policy shocks on wages, which is collinear in a specification once time fixed effects are included. As expected, monetary policy easings are associated with higher wage growth. Quantitatively, a 10 bp monetary policy easing is associated 1.5% higher wage growth. This effect is relatively stable across firms with varying degree of vacancy shares, as the model with search labor market would predict, especially as we move across columns that successively include a more extensive set of fixed effects in the regressions. Those results are also visible in Table 7, where firms that are paying a varying degree of wages are also not responding differently in terms of wage setting in response to monetary policy shocks.

Table 8 and Table 9 re-estimate columns (2) and columns (4) of Table 3 using different definitions of wages. As defined in equation (9) in section 3, equilibrium wages are measured as residuals from occupation=commuting zone level regressions and then standardized, averaged, and cumulated. For robustness, we use the pure wage measure in column (2), with ONET–time and commuting zone–time fixed effects in column (3), and ONET–time and commuting zone fixed effects in column (4). In both regression tables the results are robust to the way we measure wages. In Table 8 the wage coefficient is always negative and statistically significant, while in Table 9 the coefficients are statistically insignificant, as predicted by the model with search labor market power.

Local Projections

So far, we have only analyzed the contemporaneous effect of monetary policy. Monetary policy shocks are known to take time to affect macroeconomic variables. We employ local projection (LP) methods to confirm the finding that search monopsony is a key feature of the data.

We estimate the following LP versions of equations (10)-(13):

$$\frac{1}{H}\sum_{h=0}^{H} \mathrm{Y}_{i,c,t+h} = \alpha_H + \beta_H^{YX} \text{ MP easing}_t \times \mathrm{X}_{i,c,t-1} + \Theta^{YX}\text{controls} + \varepsilon_{i,c,t+H}^{YX} \tag{14}$$

where Y = $\log vacancies$ or $\log wages$, X = equilibrium vacancies or equilibrium wages, H is a given quarterly horizon, and i is a firm in a local labor market c at time t.[21,22]

[21]**controls** include: (i) the Federal Reserve information shock and its interactions with equilibrium vacancies and wages, (ii) equilibrium vacancies or wages depending on whether X = equilibrium vacancies or equilibrium wages are estimated, (iii) firm fixed effects that absorb any time-invariant firm variation, and (iv) commuting zone–time effects that absorb any time-varying regional shocks, such as local demand shocks.

[22]For these regressions, if Lightcast data does not report a vacancy for a given firm in a given com-

Figure 2 shows the point estimates of β_H^{YX} from equation (14) along with 95 percent confidence intervals allowing for firm and commuting–zone double clustering. The top left panel displays the interaction coefficient β^{vv}. As in the static regressions, firms with a higher share of equilibrium vacancies respond by posting more (less) vacancies following expansionary (contractionary) monetary policy, as indicated by the positive coefficient. The coefficient remains positive and statistically significant for 6 quarters after the monetary policy shock occurred.

The bottom left panel plots the interaction coefficient β^{vw}. The coefficient is persistently negative, indicating that firms that pay lower wages increase their vacancy postings by more relative to the counterparts.

The two right panels show β^{wv} (top) and β^{ww} (bottom) or the responses of wages to expansionary monetary policy shocks depending on equilibrium vacancy shares and wages, respectively. As was the case in the static regressions, these responses are economically and statistically insignificant within a six-quarter horizon.

4.3 Heterogeneity across different types of jobs

In this subsection, we investigate the effects of labor market power across job postings that require different levels of skills. Low-skilled workers are more likely to be budget constrained and thus would be more affected by longer unemployment spells making them potentially more responsive to search monopsony. (Barnichon and Figura (2015), Abbritti and Consolo (2024)) Moreover, low-skilled workers are disproportionally affected by automation, which further reduces their bargaining power (Acemoglu and Restrepo, 2022; Acemoglu et al., 2021). At the same time, low-skilled workers exhibit lower regional mobility, while skilled workers are likely to move to regions with higher wage premiums (Dahl, 2022). This creates a locked-in effect for low-skilled workers, which suggests low-skilled workers may also be more sensitive to classical monopsony. This suggests testing our predictions comparing across more and less skilled workers to assess the relative importance of search monopsony.

Lightcast provides granular data on postings, including on skill and education requirements. We focus on two types of requirements. First, we differentiate between college vs. non-college vacancies. In our sample, $\approx 40\%$ of vacancies require college education. We also study the degree of tech-savviness of vacancies, by differentiating between vacancies that require software skills and those that do not, in the spirit of Acemoglu et al. (2021) who use Lightcast data to identify AI vacancies. Vacancies that require software skills make up $\approx 28\%$ of all vacancies. As one would expect, college vacancies and tech-savvy

muting zone and quarter we assume there were no vacancies posted.

vacancies are strongly related to each other, with a correlation between the two vacancy types of $\approx 29\%$.

To test the model predictions across these types, we run the following versions of equations (10)-(11):

$$\begin{aligned} \log \mathit{vacancies}_{i,c,t,j} =& \beta^{vv} \text{MP easing}_t \times \text{equilibrium vacancies}_{i,c,t-1} + \\ & \gamma^{vv} \text{MP easing}_t \times \text{equilibrium vacancies}_{i,c,t-1} \times \text{Type}_j + \Theta^v \text{controls} + \varepsilon^v_{i,c,t} \end{aligned} \tag{15}$$

$$\begin{aligned} \log \mathit{vacancies}_{i,c,t,j} =& \beta^{vw} \text{MP easing}_t \times \text{equilibrium wages}_{i,c,t-1} + \\ & \gamma^{vw} \text{MP easing}_t \times \text{equilibrium wages}_{i,c,t-1} \times \text{Type}_j + \Theta^w \text{controls} + \varepsilon^w_{i,c,t} \end{aligned} \tag{16}$$

where Type_j is a dummy that takes the value of 0 or 1 depending on the characteristic of the job posting, and i is a firm in a local labor market c at time t.[23] The triple interaction coefficient γ captures whether there is significant heterogeneity across a particular type of vacancy. If the double interaction (β) has a different sign than the triple interaction (γ), that would mean that the effect is weaker for [Type = 1].

Table 10 shows estimates of Equation 15. Column 1 confirms our main result that vacancies respond more (less) to monetary loosening (tightening) when a firm controls a greater share of vacancies. But does this effect depend on they type of vacancy? The interaction between the monetary policy shock and the vacancy type dummy is negative and statistically significant in column 1, where the vacancy type dummy is one if the vacancy is a college vacancy. This implies that college vacancies respond less strongly to monetary policy than non-college vacancies.

Column 2 offers a more direct version of Test 2. Indeed, the interaction between the monetary policy shock, vacancy share and the vacancy type dummy is positive and statistically significant. The positive triple interaction term shows that labor demand effects of search labor market power are likely at play for non-college vacancies. When interpreting the economic magnitude, we can see that the effect is around -7.8 for non-college vacancies and $(-7.8 + 2.9) = -4.9$ for college vacancies.

The results are similar although weaker for software-related vacancies. In column 3 we can see that software vacancies are less responsive to monetary policy in general, and when firms exhibit labor market power their adjustment seems to be done along the non-software dimension, rather than on the more tech-related vacancies.

[23] **controls** include: (i) the Federal Reserve information shock and its interactions with equilibrium vacancies or wages, respectively, (ii) equilibrium vacancies or wages depending on whether equation (15) or (16) are estimated, (iii) firm fixed effects that absorb any time-invariant firm variation, (iv) commuting zone–time effects that absorb any time-varying regional shocks, such as local demand shocks, and (v) type–time fixed effects that absorb any time-varying type-specific changes, such as shifts technology benefiting skilled workers.

4.4 Employment

In this section, we ask whether employment reacts to monetary policy shocks in the same way vacancies do when the shock is interacted with our measure of labor-market power. Although our mechanism operates through vacancy postings, these postings can exaggerate true labor demand, e.g. when high inflation induces workers with sticky nominal wages to churn between jobs, inflating vacancy counts (Afrouzi et al., 2024). Re-estimating our local-projection regressions with employment as the outcome yields qualitatively identical impulse responses, dispelling concerns that our findings hinge on vacancy dynamics alone and confirming that search labor market power amplifies the effects of monetary tightening all the way to the employment margin that matters most for the macroeconomy.

Theory. The model produces the following prediction for the response of employment (and not only vacancy postings) to monetary policy shocks:

$$\hat{l}_{it} = \left(\frac{1}{1-\alpha_i} - 1\right)\hat{m}_{it} + \frac{1}{\gamma_i}\left(\frac{\alpha_i}{1-\alpha_i}\hat{m}_{it} + \hat{w}_{it} - \frac{\int_i \lambda_i \frac{1}{\gamma_i}\left(\frac{\alpha_i}{1-\alpha_i}\hat{m}_{it} + \hat{w}_{it}\right)dt}{\int_i \lambda_i \frac{1}{\gamma_i} di}\right) + \hat{N}_t$$

As in Proposition 3, if wages respond homogeneously and there is no search labor market power, labor also responds homogeneously across firms. At the same time, mirroring the results for the response of vacancies, the response of firms with higher search labor market power (high α_i) is larger than that of firms with lower search labor market power.

Compustat Dataset. We merge Lightcast and Compustat data for a large number of publicly traded firms to analyze the relationship between vacancy postings and employment growth. We aggregate vacancy postings to the firm level and fuzzy merge the Lightcast firm name to Compustat. First, we execute a standard string cleaning procedure, removing excessive spacing, acronyms related to a companys business structure acronyms, special characters, and other unnecessary characters from the name. Given the string differences between the names of companies in Lightcast and Compustat, we used a two-layered merging technique consisting of exact matching, and Jaro-Winkler string distance matching (Jaro, 1989; Winkler, 1990). For the Jaro-Winkler fuzzy matching, we set a string distance threshold of 0.11, which maximizes the number of matches and their quality jointly. We obtain 8,231 firm matches from 14,983 firms in Compustat in the years between 2008 to 2019, of which 3,217 are exact matches, and 5,014 match using the Jaro-Winkler fuzzy matching. The merged companies represent 75% of sales and 73% of employment of all companies in Compustat.

We follow the cleaning procedure of Ottonello and Winberry (2020) for the Compustat data closely. First, we keep only corporations incorporated in the US. In particular, we drop firms in the financial and utilities sectors, firm-quarter observations with acquisitions larger than 5 percent of assets, firm-quarter observations where the investment rate is in the top and bottom 1 percent of the distribution, the investment spell is shorter than 40 quarters, firm–quarter observations where liquidity, debt, and sales are outliers, firms with less than 10 million US dollars in assets, and that are in the sample less than 5 years. Since the quarterly version of Compustat does not have employment information, we linearly interpolate the annual employment data to quarterly data.

Results. To understand the relationship between vacancy postings and employment, we estimate the following equation:

$$\Delta Employment_{i,t} = \alpha_i + \alpha_t + \beta_1 \text{Log Vacancies}_{i,t} + \epsilon_{i,t} \tag{17}$$

where $\Delta Employment_{i,t}$ is the log change in employment of firm i between year t and $t-1$ in Compustat.[24] Log Vacancies$_{i,t}$ is the log number of vacancies posted by firm i in year t from Lightcast. Log Vacancies$_{i,t}$ is defined in the same way as in section 4, which allows us under certain assumptions to translate the effect of monetary policy on vacancies to an effect on employment based on the elasticity estimated in Equation 17.

Figure 2 had shown that after four quarters, a firm with high labor market power increased its vacancy postings by a factor of 2 in response to a 10 basis point accommodative monetary policy shock. A firm with medium labor market power instead did not increase its vacancy postings. Translating the vacancy postings into employment growth, we need to multiply the log number of vacancies created by the coefficient on the elasticity of employment growth to vacancies. Consequently, a firm with high labor market power has $(0.74 * 2 =)$ 1.48 percentage points stronger employment growth in response to the accommodative monetary policy shock. According to our estimates, a firm without labor market power does not exhibit stronger employment growth.

This back-of-the-envelope calculation makes several assumptions. First, we only have employment data for listed firms. For the calculation to be accurate, the elasticity needs to be the same for firms that we merge with Compustat and the firms that we do not merge. Second, the elasticity of employment growth with respect to vacancy postings may vary between firms with and without labor market power. For instance, monopsonists may post more vacancies but do not increase their actual hiring in response to an accommodative monetary policy shock, as more employees leave when labor market

[24] The contemporaneous specification reflects the fact that most vacancies are filled well before a year passes, for instance, the average time to fill a vacancy stayed at approximately one month in the time period we consider per Job Openings and Labor Turnover Survey (JOLTS).

becomes tighter in response to the shock. However, we do not find evidence in favor of a differential elasticity for firms with differential degree of labor market power, suggesting that higher vacancy postings of firms with labor market power also reflect more intense hiring and employment growth. Even for firm with a large degree of labor market power there is a strong relationship between vacancy postings and employment growth [25]

To further verify the results hold for employment rather than vacancies, we estimate the dynamic response of employment in Compustat through the following:

$$\sum_{h=0}^{H} \Delta\text{Employment}_{i,t+h} = \alpha_H + \beta_H \text{ MP easing}_t \times \text{LMP}_{i,t-1} + \theta_H X_{i,t} + \gamma_i + \gamma_{t+H} + \varepsilon_{i,t+H}$$

where $\Delta\text{Employment}_{i,t+h}$ is defined as the log difference between employment in time t+h and t-1. MP easing_t is the negative of the monetary policy shock by Jarociński and Karadi (2020), in which a positive value reflects monetary policy easing. $\text{LMP}_{i,t-1}$ is defined as one if the firm is in the top 5% of labor market power in at least one commuting zone. Standard errors are clustered at the firm level. Shaded areas represent a 95% confidence interval.

Figure 3 plots β_h of this regression. In response to a 10 bp monetary easing shock firms with labor market power increase their employment by 1-2 percentage point more than firms without labor market power within the first year. The result is consistent with the above back-of-the-envelope calculation, under which the response is around 1.5%.

5 Conclusion

Where does labor market power come from and why does it matter? A large body of work has focused on monopsony power that derives from imperfect substitution between labor markets, for example because jobs are closer to where workers live. Under this type of "classical" monopsony, wages are driven below the marginal revenue of labor inefficiently and employment is too low. That deadweight loss creates a rationale for several policies, like minimum wages. Another strand of the literature studies a different type of monopsony power that arises due to search frictions. Because applicants face uncertain job prospects and that uncertainty is unwarranted, firms that are better at converting applicants into employees derive an advantage that can be seen as monopsony power too. Unlike classical monopsony, such "search" monopsony can be efficient or even if inefficient, generate excess employment. Minimum wages may be less compelling if search monopsony is pervasive.

[25] The positive relationship is robust to using other specifications, such as log-log equations.

To the best of our knowledge this paper is the first to include the two types of monopsony power in a simple framework and to test their relative pervasiveness. To do so, we first derive predictions for equilibrium vacancies and wages and responses of vacancies and wages to demand shocks in a model that is rich in “classical” monopsony with simplified directed search elements. These predictions draw a sharp distinction between classical and search monopsony.

We test these predictions by using the near universe of online vacancy postings in the U.S. and find that the cross-section of wages and vacancies is better rationalized by the presence of search monopsony, implying a back-of-the-envelope calculation that at least 55% of markdown cross-firm variation is driven by cross-firm variation in search labor market power. Moreover, we also find that the response of vacancies and wages to demand shocks measured by exogenous monetary policy shocks can only be explained with a preponderance of search rather than classical monopsony power.

The preponderance of search monopsony implies that markdowns may be at least partly efficient. Thus, we hope that our paper will spur further quantitative research in distinguishing these two forms of monopsony power and the optimality of labor market policies, such as minimum wages.

References

Abbritti, M. and A. Consolo (2024) "Labour market skills, endogenous productivity and business cycles", *European Economic Review*, 170, p. 104873. 12, 30

Acemoglu, D. and P. Restrepo (2022) "Tasks, automation, and the rise in us wage inequality", *Econometrica*, 90 (5), pp. 1973–2016. 30

Acemoglu, Daron, Joe Hazell, and Pascual Restrepo (2021) "Ai and jobs: evidence from online vacancies", *Journal of Labor Economics*. 30

Afrouzi, Hassan, Andres Blanco, Andres Drenik, and Erik Hurst (2024) "A theory of how workers keep up with inflation",Technical report, National Bureau of Economic Research. 32

Andersen, Asger Lau, Niels Johannesen, Mia Jørgensen, and José-Luis Peydró (2021) "Monetary policy and inequality". 6

Azar, José, Steven Berry, and Ioana Elena Marinescu (2019a) "Estimating labor market power", *Available at SSRN 3456277*. 6

Azar, José, Emiliano Huet-Vaughn, Ioana Marinescu, Bledi Taska, and Till Von Wachter (2019b) "Minimum wage employment effects and labor market concentration",Technical report, National Bureau of Economic Research. 6

Azar, José, Ioana Marinescu, and Marshall Steinbaum (2019c) "Measuring labor market power two ways", *AEA Papers and Proceedings*, 109, pp. 317–21. 6

Azar, José, Ioana Marinescu, and Marshall Steinbaum (2022) "Labor market concentration", *Journal of Human Resources*, 57 (S), pp. S167–S199. 6, 7

Azar, José, Ioana Marinescu, Marshall Steinbaum, and Bledi Taska (2020) "Concentration in us labor markets: Evidence from online vacancy data", *Labour Economics*, 66, p. 101886. 6

Barnichon, R. and A. Figura (2015) "Labor market heterogeneity and the aggregate matching function", *American Economic Journal: Macroeconomics*, 7 (4), pp. 222–249. 12, 30

Bartscher, Alina K, Moritz Kuhn, Moritz Schularick, and Paul Wachtel (2021) "Monetary policy and racial inequality". 6

Bauer, Michael D. and Eric T. Swanson (2023) "An Alternative Explanation for the "Fed Information Effect"", *American Economic Review*, 113 (3), pp. 664–700. 23

Benmelech, Efraim, Nittai K Bergman, and Hyunseob Kim (2022) "Strong employers and weak employees how does employer concentration affect wages?", *Journal of Human Resources*, 57 (S), pp. S200–S250. 6

Berger, D., K. Herkenhoff, A. R. Kostøl, and S. Mongey (2024) "An anatomy of monopsony: Search frictions, amenities, and bargaining in concentrated markets", *NBER Macroeconomics Annual*, 38 (1), pp. 1–47. 6, 15, 17

Berger, David, Kyle Herkenhoff, and Simon Mongey (2022) "Labor market power", *American Economic Review*, 112 (4), pp. 1147–93. 2, 3, 6, 7, 8, 9, 11, 15, 25

Bergman, Nittai, David A Matsa, and Michael Weber (2022) "Inclusive monetary policy: How tight labor markets facilitate broad-based employment growth",Technical report, National Bureau of Economic Research. 6

Bhaskar, Venkataraman, Alan Manning, and Ted To (2002) "Oligopsony and monopsonistic competition in labor markets", *Journal of Economic Perspectives*, 16 (2), pp. 155–174. 2

Bu, Chunya, John Rogers, and Wenbin Wu (2021) "A unified measure of fed monetary policy shocks", *Journal of Monetary Economics*, 118, pp. 331–349. 23, 51, 52

Burdett, Kenneth and Dale T. Mortensen (1998) "Wage differentials, employer size, and unemployment", *International Economic Review*, pp. 257–273. 2, 6

Burdett, Kenneth, Shouyong Shi, and Randall Wright (2001) "Pricing and matching with frictions", *Journal of Political Economy*, 109 (5), pp. 1060–1085. 6

Burya, Anastasia, Rui Mano, Yannick Timmer, and Anke Weber (2023) "The wage phillips curve under labor market power", *AEA Papers and Proceedings*, 113, pp. 110–113, American Economic Association 2014 Broadway, Suite 305, Nashville, TN 37203. 5

Campbell, Jeffrey R, Charles L Evans, Jonas DM Fisher, Alejandro Justiniano, Charles W Calomiris, and Michael Woodford (2012) "Macroeconomic effects of federal reserve forward guidance", *Brookings papers on economic activity*, pp. 1–80. 23

Card, David, Ana Rute Cardoso, Joerg Heining, and Patrick Kline (2018) "Firms and labor market inequality: Evidence and some theory", *Journal of Labor Economics*, 36 (S1), pp. S13–S70. 11, 15

Coglianese, John, Maria Olsson, and Christina Patterson (2021) "Monetary policy and the labor market: a quasi-experiment in sweden",Technical report, Working Paper. 6

Coibion, Olivier, Yuriy Gorodnichenko, Lorenz Kueng, and John Silvia (2017) "Innocent bystanders? monetary policy and inequality", *Journal of Monetary Economics*, 88, pp. 70–89. 6

Dahl, G. B. (2022) "Mobility and the return to education: Testing a roy model with multiple markets", *Econometrica*, 70 (6), pp. 2367–2420. 30

Davis, S. J., Faberman R. J., and J. C. Haltiwanger (2013) "The establishment-level behavior of vacancies and hiring", *The Quarterly Journal of Economics*, 128 (2), pp. 581–622. 6, 12

De Loecker, Jan, Jan Eeckhout, and Gabriel Unger (2020) "The rise of market power and the macroeconomic implications", *The Quarterly Journal of Economics*, 135 (2), pp. 561–644. 46

Dolado, Juan J, Gergő Motyovszki, and Evi Pappa (2021) "Monetary policy and inequality under labor market frictions and capital-skill complementarity", *American economic journal: macroeconomics*, 13 (2), pp. 292–332. 6

Fornaro, Luca and Martin Wolf (2021) "Monetary policy in the age of automation". 6

Grigsby, John, Erik Hurst, and Ahu Yildirmaz (2021) "Aggregate nominal wage adjustments: New evidence from administrative payroll data", *American Economic Review*, 111 (2), p. 428471. 19

Haefke, Christian and Marcus Sonntag (2013) "Wage rigidity and job creation", *Journal of Monetary Economics*, 60 (8), pp. 887–899. 19

Hazell, Jonathon, Christina Patterson, H Sarsons, and B Taska (2021) "National wage setting",Technical report, Working Paper. 7, 21, 28

Hazell, Jonathon and Bledi Taska (2020) "Downward rigidity in the wage for new hires", (3728939). 19

Hershbein, Brad and Lisa B Kahn (2018) "Do recessions accelerate routine-biased technological change? evidence from vacancy postings", *American Economic Review*, 108 (7), pp. 1737–72. 7, 21

Hershbein, Brad, Claudia Macaluso, and Chen Yeh (2022) "Concentration in us local labor markets: evidence from vacancy and employment data", *American Economic Review (forthcoming)*. 2, 6, 7

Hosios, Arthur J. (1990) "On the efficiency of matching and related models of search and unemployment", *Review of Economic Studies* (2), pp. 279–298. 2

Jaro, Matthew A (1989) "Advances in record-linkage methodology as applied to matching the 1985 census of tampa, florida", *Journal of the American Statistical Association*, 84 (406), pp. 414–420. 32

Jarocinski, Marek (2021) "Estimating fed's unconventional policy shocks". 23, 51, 52

Jarociński, Marek and Peter Karadi (2020) "Deconstructing monetary policy surprisesthe role of information shocks", *American Economic Journal: Macroeconomics*, 12 (2), pp. 1–43. 4, 22, 34, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52

Jarosch, Gregor, Jan Sebastian Nimczik, and Isaac Sorkin (2019) "Granular search, market structure, and wages", *National Bureau of Economic Research Working Paper No. w26239*. 2, 6

Jasova, Martina, Caterina Mendicino, Ettore Panetti, José-Luis Peydró, and Dominik Supera (2021) "Monetary policy, labor income redistribution and the credit channel: Evidence from matched employer-employee and credit registers". 6

Lange, Fabian and Theodore Papageorgiou (2020) "Beyond cobb-douglas: flexibly estimating matching functions with unobserved matching efficiency", *NBER Working Papers*, No. w26972. 6, 12

Lerner, Abba P. (1934) "The concept of monopoly and the measurement of monopoly power", *Review of Economic Studies*, 1 (3), p. 157175. 10

Manning, Alan. (2021) "Monopsony in labor markets: A review", *ILR Review*, 74 (1), pp. 3–26. 15

Martins, Pedro S., Gary Solon, and Jonathan P. Thomas (2012) "Measuring what employers do about entry wages over the business cycle: A new approach", *American Economic Journal: Macroeconomics*, 4 (4), pp. 36–55. 19

Moen, Espen R. (1997) "Competitive search equilibrium", *Journal of Political Economy* (2), pp. 385–411. 2, 3, 6, 7, 11

Nakamura, Emi and Jón Steinsson (2018) "High-frequency identification of monetary non-neutrality: the information effect", *The Quarterly Journal of Economics*, 133 (3), pp. 1283–1330. 23, 51, 52

Ottonello, Pablo and Thomas Winberry (2020) "Financial heterogeneity and the investment channel of monetary policy", *Econometrica*, 88 (6), pp. 2473–2502. 32

Pissarides, Christopher A. (1985) "Short-run equilibrium dynamics of unemployment, vacancies, and real wages", *American Economic Review* (4), pp. 676–690. 2, 6

Robinson, Joan L. (1969) *The Economics of Imperfect Competition*, Macmillan. 2

Romer, Christina D and David H Romer (1989) "Does monetary policy matter? a new test in the spirit of friedman and schwartz", *NBER macroeconomics annual*, 4, pp. 121–170. 6

Romer, Christina D. and David H. Romer (2000) "Federal reserve information and the behavior of interest rates", *American Economic Review*, 90 (3), p. 429457. 23

Rudanko, Leena (2023) "Firm wages in a frictional labor market", *American Economic Journal: Macroeconomics* (1), pp. 517–550. 2, 6

Winkler, William E (1990) "String comparator metrics and enhanced decision rules in the fellegi-sunter model of record linkage.". 32

Wolthoff, R. (2018) "Applications and interviews: Firms recruiting decisions in a frictional labour market", *The Review of Economic Studies*, 85 (2), pp. 1314–1351. 6, 12

Table 1: Summary of the Panel Structure

	Total	Firms	Commuting Zone	Time
Number of Observations	15,810,352	387,107	708	43
Average Number of Firms	387,107	-	22,412	103,230
Average Number of CZ	708	170	-	704
Average Number of Periods	43	29	42	-

This table reports the total number of observations and the number of observations across the time and geographical dimensions of the data

Table 2: Equilibrium Wages and Vacancies (Test 1)

	Log wage$_{v,i,c,t}$				
	(1)	(2)	(3)	(4)	(5)
Equilibirum Vacancy Share	-0.232***	-0.232***	-0.175***	-0.188***	-0.193***
	(-3.84)	(-3.84)	(-3.91)	(-3.49)	(-4.14)
College$_{v,i,c,t}$	0.238***	0.238***	0.172***	0.170***	0.162***
	(41.47)	(41.47)	(42.51)	(43.17)	(45.50)
Software Skills$_{v,i,c,t}$	0.028***	0.028***	0.016***	0.015***	0.014***
	(8.13)	(8.13)	(5.09)	(5.01)	(5.08)
Specialized$_{v,i,c,t}$	0.088***	0.088***	0.065***	0.065***	0.063***
	(29.75)	(29.75)	(25.62)	(27.11)	(28.45)
Routine Manual$_{v,i,c,t}$	-0.110***	-0.110***	0.038***	0.038***	0.050***
	(-18.86)	(-18.86)	(4.20)	(4.31)	(5.97)
Routine Cognitive$_{v,i,c,t}$	-0.144***	-0.144***	0.061***	0.062***	0.077***
	(-28.17)	(-28.17)	(6.99)	(7.72)	(9.46)
Non-Routine Manual Physical$_{v,i,c,t}$	-0.057***	-0.057***	0.040***	0.045***	0.039***
	(-9.41)	(-9.41)	(4.48)	(5.12)	(5.27)
Non-Routine Manual Inter-Personal$_{v,i,c,t}$	0.045***	0.045***	0.035***	0.035***	0.039***
	(10.42)	(10.42)	(4.81)	(4.91)	(5.01)
Non-Routine Cognitive Analytical$_{v,i,c,t}$	0.064***	0.064***	0.111***	0.113***	0.112***
	(12.24)	(12.24)	(16.58)	(17.61)	(16.69)
Non-Routine Cognitive Personal$_{v,i,c,t}$	0.159***	0.159***	0.045***	0.046***	0.046***
	(33.83)	(33.83)	(8.67)	(9.32)	(10.02)
Obs.	12,737,416	12,737,416	12,237,549	12,233,118	12,232,649
Firm FE	✓	✓	✓	✓	✓
CZ × Time FE	✓	✓	✓	✓	✓
Industry × Time FE	✓	✓	✓	✓	✓
ONET × Time FE			✓	✓	✓
ONET × CZ FE				✓	✓
ONET × Industry FE					✓
No. Firms	174,270	174,270	171,560	171,524	171,521

This table reports results for the following vacancy-level regression: Log wage$_{v,i,c,t} = \alpha + \beta$ Equilibrium Vacancies$_{i,c,t}$ + $\theta X_{v,i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \gamma_{ind,t} + \gamma_{occ,t} + \gamma_{occ,c} + \gamma_{occ,ind} + \varepsilon_{v,i,c,t}$, where Log wages$_{v,i,c,t}$ is defined as the log of posted wage in vacancy v for firm i, in commuting zone c in quarter t. Equilibrium Vacancy Share$_{i,c,t}$ is defined as the cumulative vacancy share of firm i in commuting zone c at quarter t. $X_{v,i,ct}$ is a vector of vacancy characteristics as defined in section 3. $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time, and $\gamma_{ind,t}$ are industry-time, $\gamma_{occ,t}$ are occupation-time, $\gamma_{occ,c}$ are occupation-CZ, and $\gamma_{occ,ind}$ are occupation-industry fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Table 3: Monetary Policy under Labor Market Power (Test 2)

	Log Vacancies$_{i,c,t}$		Log Wages$_{i,c,t}$	
	(1)	(2)	(3)	(4)
MP easing$_t$ × Equilibrium Vacancies	6.906**		0.128	
	(2.707)		(0.377)	
MP easing$_t$ × Equilibrium Wages		-0.061**		-0.020
		(0.028)		(0.033)
Obs.	15,069,930	4,135,037	3,545,581	1,827,037
Firm FE	✓	✓	✓	✓
CZ × Time FE	✓	✓	✓	✓
No. Firms	355,254	145,726	216,310	110,303

This table reports estimates of the following regression: $Y_{i,c,t} = \alpha + \beta$ MP easing$_t \times X_{i,c,t-1} + \theta Z_{i,c,t} + \gamma_i + \gamma_{c,t} + \varepsilon_{i,c,t}$, where $Y_{i,c,t}$ is the dependent variable (Log Vacancies$_{i,c,t}$, Log Wages$_{i,c,t}$). Log Vacancies$_{i,c,t}$ is defined as the log number of vacancies posted by firm i, in commuting zone c in quarter t. Log Wages$_{i,c,t}$ is defined as the log deviation of the firm i wage in the commuting zone c at the time t from the average wage in the corresponding commuting zone. MP$_t$ is the negative of the monetary policy shock by Jarociński and Karadi (2020), and thus a positive value reflects monetary policy easing. $X_{i,c,t}$ is the equilibrium wage or equilibrium vacancy measure. Equilibrium vacancy measure is defined as the cumulative vacancy share of firm i in commuting zone c at quarter $t-1$. For more details see section 3. Equilibrium wage measure is defined as the standartized average cumulative residual of the ONET-level regression. The ONET-level regression is given by Wage$_{i,c,n,t} = \alpha + \gamma_{n,t} + \gamma_{c,t}$, where Wage is the wage of the firm i in the commuting zone c at the time t for ONET occupation n; $\gamma_{n,t}$ are ONET-time fixed effects and $\gamma_{c,t}$ are CZ-time fixed effects. γ_i is firm-level fixed effects, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Table 4: Labor Demand Effect of Monetary Policy - Vacancies to Vacancies

	Log Vacancies$_{i,c,t}$				
	(1)	(2)	(3)	(4)	(5)
MP easing$_t$	0.696***				
	(0.035)				
Equilibrium Vacancies	20.318***	14.958***	20.866***	22.265***	22.713***
	(1.536)	(1.277)	(1.563)	(1.569)	(1.655)
MP easing$_t$ × Equilibrium Vacancies	5.442**	5.439***	7.624***	6.906**	7.895**
	(2.329)	(1.833)	(2.397)	(2.711)	(3.862)
Obs.	15,070,026	15,070,026	15,070,026	15,069,930	12,851,727
Firm FE	✓	✓	✓	✓	
CZ FE	✓		✓		
Time FE		✓	✓		
CZ × Time FE				✓	✓
Firm × Time FE					✓
No. Firms	355,254	355,254	355,254	355,254	199,839

This table reports estimates of the following regression: Log Vacancies$_{i,c,t} = \alpha + \beta$ MP$_t \times$ Eq.Vac$_{i,c,t-1} + \theta Z_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where Log Vacancies$_{i,c,t}$ is defined as the log number of vacancies posted by firm i, in commuting zone c in quarter t. MP$_t$ is the negative of the monetary policy shock by Jarociński and Karadi (2020), and thus a positive value reflects monetary policy easing. Eq.Vac$_{i,c,t-1}$ is defined as the cumulative vacancy share of firm i in commuting zone c at quarter $t-1$. For more details see section 3. $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Table 5: Labor Demand Effect of Monetary Policy - Vacancies to Wages

	Log Vacancies$_{i,c,t}$				
	(1)	(2)	(3)	(4)	(5)
MP easing$_t$	0.874***				
	(0.057)				
Equilibrium Wages	0.015***	-0.004	0.013***	0.013***	0.013***
	(0.004)	(0.005)	(0.004)	(0.004)	(0.004)
MP easing$_t$ × Equilibrium Wages	-0.058**	-0.067**	-0.062**	-0.061**	0.008
	(0.028)	(0.027)	(0.028)	(0.028)	(0.025)
Obs.	4,135,855	4,135,856	4,135,855	4,135,037	3,295,071
Firm FE	✓	✓	✓	✓	
CZ FE	✓		✓		
Time FE		✓	✓		
CZ × Time FE				✓	✓
Firm × Time FE					✓
No. Firms	145,726	145,726	145,726	145,721	47,267

This table reports estimates of the following regression: Log Vacancies$_{i,c,t} = \alpha + \beta$ MP$_t \times$ Eq.Wage$_{i,c,t-1} + \theta Z_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where Log Vacancies$_{i,c,t}$ is defined as the log number of vacancies posted by firm i, in commuting zone c in quarter t. MP$_t$ is the negative of the monetary policy shock by Jarociński and Karadi (2020), and thus a positive value reflects monetary policy easing. Eq.Wage$_{i,c,t-1}$ is defined as the standartized average cumulative residual of the ONET-level regression. The ONET-level regression is given by Wage$_{i,c,n,t} = \alpha + \gamma_{n,t} + \gamma_{c,t}$, where Wage is the wage of the firm i in the commuting zone c at the time t for ONET occupation n; $\gamma_{n,t}$ are ONET-time fixed effects and $\gamma_{c,t}$ are CZ-time fixed effects. For more details see section 3. $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Table 6: Labor Demand Effect of Monetary Policy - Wages to Vacancies

	Log Wages$_{i,c,t}$				
	(1)	(2)	(3)	(4)	(5)
MP easing$_t$	0.148***				
	(0.024)				
Equilibrium Vacancies	0.056	-0.011	0.112*	0.108	0.390***
	(0.061)	(0.093)	(0.065)	(0.077)	(0.081)
MP easing$_t$ × Equilibrium Vacancies	-0.495*	0.009	0.090	0.128	0.363
	(0.279)	(0.271)	(0.277)	(0.378)	(0.485)
Obs.	3,546,366	3,546,366	3,546,366	3,545,581	2,715,673
Firm FE	✓	✓	✓	✓	
CZ FE	✓		✓		
Time FE		✓	✓		
CZ × Time FE				✓	✓
Firm × Time FE					✓
No. Firms	216,315	216,315	216,315	216,310	97,856

This table reports estimates of the following regression: Log Wages$_{i,c,t} = \alpha + \beta$ MP$_t \times$ Eq.Vac$_{i,c,t-1} + \theta Z_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where Log Wages$_{i,c,t}$ is defined as the log deviation of the firm i wage in the commuting zone c at the time t from the average wage in the corresponding commuting zone. MP$_t$ is the negative of the monetary policy shock by Jarociński and Karadi (2020), and thus a positive value reflects monetary policy easing. Eq.Vac$_{i,c,t-1}$ is defined as the cumulative vacancy share of firm i in commuting zone c at quarter $t-1$. For more details see section 3. $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Table 7: Labor Demand Effect of Monetary Policy - Wages to Wages

	Log Wages$_{i,c,t}$				
	(1)	(2)	(3)	(4)	(5)
MP easing$_t$	0.161***				
	(0.027)				
Equilibrium Wages	0.134***	0.136***	0.133***	0.134***	0.123***
	(0.004)	(0.004)	(0.004)	(0.004)	(0.003)
MP easing$_t$ × Equilibrium Wages	-0.021	-0.012	-0.013	-0.020	0.016
	(0.032)	(0.032)	(0.033)	(0.033)	(0.031)
Obs.	1,828,640	1,828,641	1,828,640	1,827,037	1,359,325
Firm FE	✓	✓	✓	✓	
CZ FE	✓		✓		
Time FE		✓	✓		
CZ × Time FE				✓	✓
Firm × Time FE					✓
No. Firms	110,318	110,318	110,318	110,303	36,606

This table reports estimates of the following regression: Log Wages$_{i,c,t}$ $= \alpha + \beta$ MP$_t \times$ Eq.Vac$_{i,c,t-1} + \theta Z_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where Log Wages$_{i,c,t}$ is defined as the log deviation of the firm i wage in the commuting zone c at the time t from the average wage in the corresponding commuting zone. MP$_t$ is the negative of the monetary policy shock by Jarociński and Karadi (2020), and thus a positive value reflects monetary policy easing. Eq.Wage$_{i,c,t-1}$ is defined as the standartized average cumulative residual of the ONET-level regression. The ONET-level regression is given by Wage$_{i,c,n,t}$ $= \alpha + \gamma_{n,t} + \gamma_{c,t}$, where Wage is the wage of the firm i in the commuting zone c at the time t for ONET occupation n; $\gamma_{n,t}$ are ONET-time fixed effects and $\gamma_{c,t}$ are CZ-time fixed effects. For more details see section 3. $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Table 8: Labor Demand Effect of Monetary Policy - Vacancies to Wages (other measures)

	Log Vacancies$_{i,c,t}$			
	(1)	(2)	(3)	(4)
Equilibrium Wages	0.013***	0.054***	0.009**	0.014***
	(0.004)	(0.001)	(0.004)	(0.004)
MP easing$_t$ × Equilibrium Wages	-0.061**	-0.126***	-0.056**	-0.060**
	(0.028)	(0.017)	(0.028)	(0.028)
Obs.	4,135,037	6,133,843	4,134,099	4,135,527
Firm FE	✓	✓	✓	✓
CZ × Time FE	✓	✓	✓	✓
No. Firms	145,721	188,583	145,712	145,725
Residuals fe				
ONET × Time FE	✓		✓	✓
CZ × Time FE	✓			
CZ FE				✓
CZ × ONET FE			✓	

This table reports estimates of the following regression: Log Vacancies$_{i,c,t}$ $= \alpha + \beta$ MP$_t \times$ Eq.Wage$_{i,c,t-1} + \theta Z_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where Log Vacancies$_{i,c,t}$ is defined as the log number of vacancies posted by firm i, in commuting zone c in quarter t. MP$_t$ is the negative of the monetary policy shock by Jarociński and Karadi (2020), and thus a positive value reflects monetary policy easing. Eq.Wage$_{i,c,t-1}$ is defined differently in each column to access robustness. For columns 1, 5 and 6, Eq.Wage$_{i,c,t-1}$ is defined as the standartized average cumulative residual of the ONET-level regression. The ONET-level regression is given by Wage$_{i,c,n,t}$ $= \alpha + \gamma_{n,t} + \gamma_{c,t}$, where Wage is the wage of the firm i in the commuting zone c at the time t for ONET occupation n; included fixed effects are, respectively, ONET-time and commuting-zone(CZ)-time, ONET-time and commuting-zone(CZ)-ONET, ONET-time and commuting-zone(CZ). For column 3 the Eq.Wage$_{i,c,t-1}$ is defined as log of the overtime average salary for the firm i in commuting zone c at the quarter t. For column 4 the Eq.Wage$_{i,c,t-1}$ is defined as a cumulative average wage. For more details see section 3. $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Table 9: Labor Demand Effect of Monetary Policy - Wages to Wages (other measures)

	Log $\text{Wages}_{i,c,t}$			
	(1)	(2)	(3)	(4)
Equilibrium Wages	0.134***	0.881***	0.125***	0.135***
	(0.004)	(0.003)	(0.004)	(0.004)
MP easing_t × Equilibrium Wages	-0.020	-0.007	-0.025	-0.020
	(0.033)	(0.044)	(0.034)	(0.033)
Obs.	1,827,037	3,577,157	1,826,927	1,827,072
Firm FE	✓	✓	✓	✓
CZ × Time FE	✓	✓	✓	✓
No. Firms	110,303	216,780	110,298	110,303
Residuals fe				
ONET × Time FE	✓		✓	✓
CZ × Time FE	✓			
CZ				✓
CZ × ONET FE			✓	

This table reports estimates of the following regression: Log $\text{Wages}_{i,c,t} = \alpha + \beta\ \text{MP}_t \times \text{Eq.Vac}_{i,c,t-1} + \theta Z_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where Log $\text{Wages}_{i,c,t}$ is defined as the log deviation of the firm i wage in the commuting zone c at the time t from the average wage in the corresponding commuting zone. MP_t is the negative of the monetary policy shock by Jarociński and Karadi (2020), and thus a positive value reflects monetary policy easing. $\text{Eq.Wage}_{i,c,t-1}$ is defined differently in each column to access robustness. For columns 1, 5 and 6, $\text{Eq.Wage}_{i,c,t-1}$ is defined as the standartized average cumulative residual of the ONET-level regression. The ONET-level regression is given by $\text{Wage}_{i,c,n,t} = \alpha + \gamma_{n,t} + \gamma_{c,t}$, where Wage is the wage of the firm i in the commuting zone c at the time t for ONET occupation n; included fixed effects are, respectively, ONET-time and commuting-zone(CZ)-time, ONET-time and commuting-zone(CZ)-ONET, ONET-time and commuting-zone(CZ). For column 3 the $\text{Eq.Wage}_{i,c,t-1}$ is defined as log of the overtime average salary for the firm i in commuting zone c at the quarter t. For column 4 the $\text{Eq.Wage}_{i,c,t-1}$ is defined as a cumulative average wage. For more details see section 3. $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Table 10: Labor Demand Effect of Monetary Policy - Heterogeneity to Vacancies

	Log $\text{Vacancies}_{i,c,t,j}$			
	(1)	(2)	(3)	(4)
Equilibrium $\text{Vacancies}_{i,c,t}$	17.417***	18.600***	17.751***	21.119***
	(1.202)	(1.270)	(1.220)	(1.434)
Type	-0.132***		-0.204***	
	(0.016)		(0.013)	
MP easing_t × Equilibrium $\text{Vacancies}_{i,c,t}$	5.056**	6.603***	5.144***	6.116**
	(1.986)	(2.120)	(1.922)	(2.922)
MP easing_t × Type	-0.343***		-0.163***	
	(0.043)		(0.041)	
Equilibrium $\text{Vacancies}_{i,c,t}$ × Type		-2.423***		-8.017***
		(0.561)		(0.711)
MP easing_t × Equilibrium $\text{Vacancies}_{i,c,t}$ × Type		-3.305**		-3.133
		(1.635)		(2.527)
Obs.	19,007,773	19,007,773	17,956,074	17,956,074
Vacancy Type	college	college	software	software
Firm FE	✓	✓	✓	✓
CZ× Time FE	✓	✓	✓	✓
Vac. Type× Time FE		✓		✓

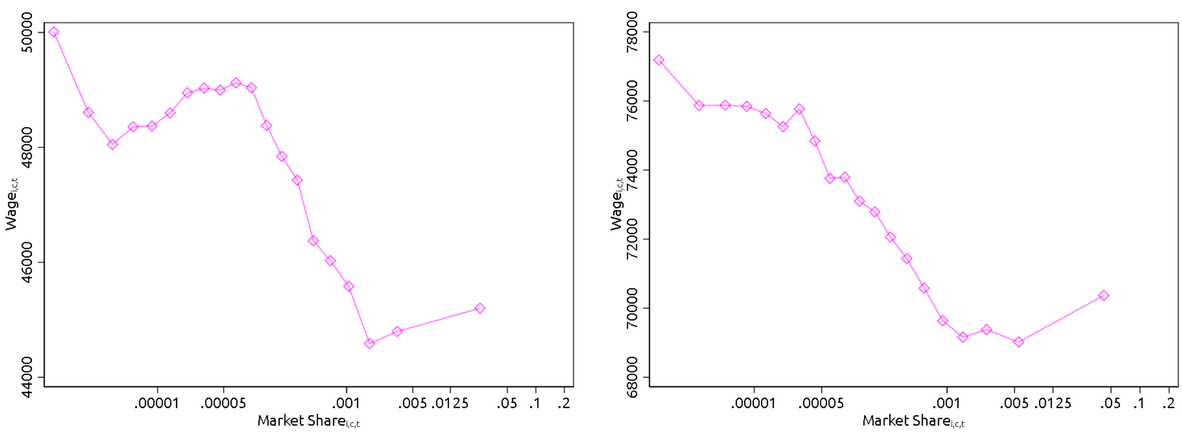
This table reports estimates of the following regression: Log $\text{Wages}_{i,c,t} = \alpha + \beta\ \text{MP}_t \times \text{Eq.Vac}_{i,c,t-1} + \theta Z_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where Log $\text{Wages}_{i,c,t}$ is defined as the log deviation of the firm i wage in the commuting zone c at the time t from the average wage in the corresponding commuting zone. MP_t is the negative of the monetary policy shock by Jarociński and Karadi (2020), and thus a positive value reflects monetary policy easing. $\text{Eq.Wage}_{i,c,t-1}$ is defined differently in each column to access robustness. For columns 1, 5 and 6, $\text{Eq.Wage}_{i,c,t-1}$ is defined as the standartized average cumulative residual of the ONET-level regression. The ONET-level regression is given by $\text{Wage}_{i,c,n,t} = \alpha + \gamma_{n,t} + \gamma_{c,t}$, where Wage is the wage of the firm i in the commuting zone c at the time t for ONET occupation n; included fixed effects are, respectively, ONET-time and commuting-zone(CZ)-time, ONET-time and commuting-zone(CZ)-ONET, ONET-time and commuting-zone(CZ). For column 3 the $\text{Eq.Wage}_{i,c,t-1}$ is defined as log of the overtime average salary for the firm i in commuting zone c at the quarter t. For column 4 the $\text{Eq.Wage}_{i,c,t-1}$ is defined as a cumulative average wage. For more details see section 3. $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Table 11: Labor Demand Effect of Monetary Policy - Heterogeneity to Wages

	Log Vacancies$_{i,c,t,j}$			
	(1)	(2)	(3)	(4)
Equilibrium Wages	0.075***	-0.059***	0.107***	0.020**
	(0.007)	(0.011)	(0.008)	(0.009)
Type	-0.096***		-0.222***	
	(0.019)		(0.014)	
MP easing$_t$ × Equilibrium Wages	-0.156***	-0.218***	-0.126***	-0.161***
	(0.048)	(0.077)	(0.044)	(0.055)
MP easing$_t$ × Type	-0.182***		-0.173***	
	(0.053)		(0.043)	
Equilibrium Wages × Type		0.301***		0.275***
		(0.021)		(0.018)
MP easing$_t$ × Equilibrium Wages × Type		0.121		0.143*
		(0.086)		(0.081)
Obs.	11,082,029	11,082,029	10,588,514	10,588,514
Vacancy Type	college	college	software	software
Firm FE	✓	✓	✓	✓
CZ× Time FE	✓	✓	✓	✓
Vac. Type× Time FE		✓		✓

This table reports estimates of the following regression: Log Wages$_{i,c,t} = \alpha + \beta\ \text{MP}_t \times \text{Eq.Vac}_{i,c,t-1} + \theta Z_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where Log Wages$_{i,c,t}$ is defined as the log deviation of the firm i wage in the commuting zone c at the time t from the average wage in the corresponding commuting zone. MP$_t$ is the negative of the monetary policy shock by Jarociński and Karadi (2020), and thus a positive value reflects monetary policy easing. Eq.Wage$_{i,c,t-1}$ is defined differently in each column to access robustness. For columns 1, 5 and 6, Eq.Wage$_{i,c,t-1}$ is defined as the standartized average cumulative residual of the ONET-level regression. The ONET-level regression is given by Wage$_{i,c,n,t} = \alpha + \gamma_{n,t} + \gamma_{c,t}$, where Wage is the wage of the firm i in the commuting zone c at the time t for ONET occupation n; included fixed effects are, respectively, ONET-time and commuting-zone(CZ)-time, ONET-time and commuting-zone(CZ)-ONET, ONET-time and commuting-zone(CZ). For column 3 the Eq.Wage$_{i,c,t-1}$ is defined as log of the overtime average salary for the firm i in commuting zone c at the quarter t. For column 4 the Eq.Wage$_{i,c,t-1}$ is defined as a cumulative average wage. For more details see section 3. $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Figure 1: Vacancy Share and Wages

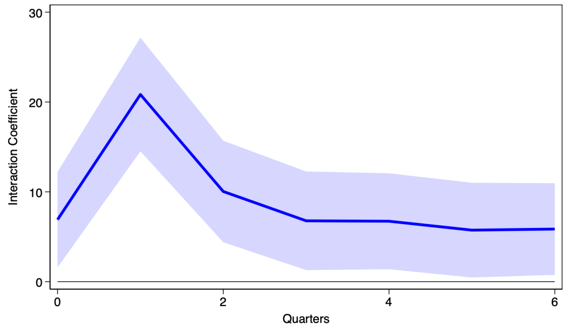


(a) Vacancies *without* a college requirement

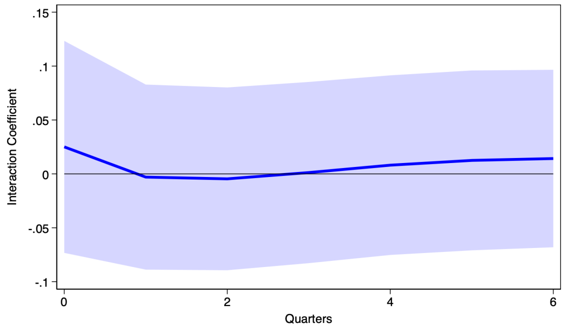
(b) Vacancies *with* a college requirement

This figure plots a local polynomial smooth of wages on vacancy share for non-college (left panel) and college (right panel) vacancies. The wages are defined as the average wage posted by firm i in commuting zone c in quarter t. The vacancy share defined as Vacancy Share$_{i,c,t} = \frac{\sum_{\tau \leq t} v_{i,c\tau}}{\sum_{\tau \leq t} \sum_i v_{i,c\tau}}$ for each firm i in commuting zone c in quarter t.

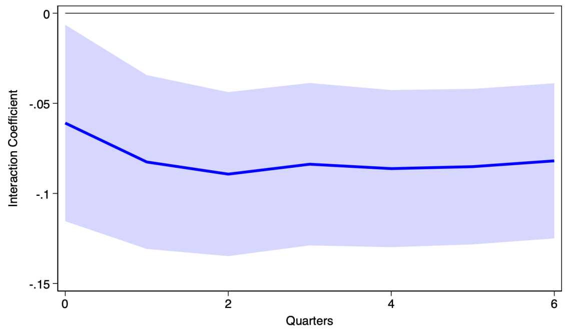
Figure 2: Cumulative Impulse Response of Vacancies and Wages to Monetary Policy



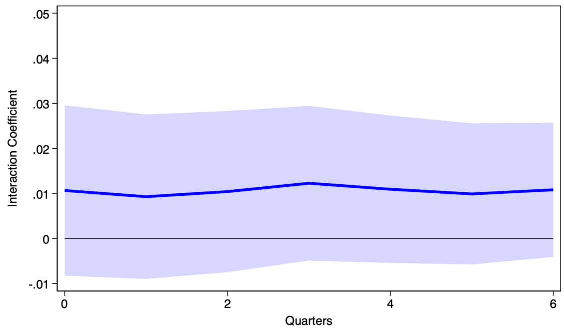
(a) Elasticity of vacancies to vacancies, $\hat{\beta}^{vv}$



(b) Elasticity of wages to vacancies, $\hat{\beta}^{wv}$



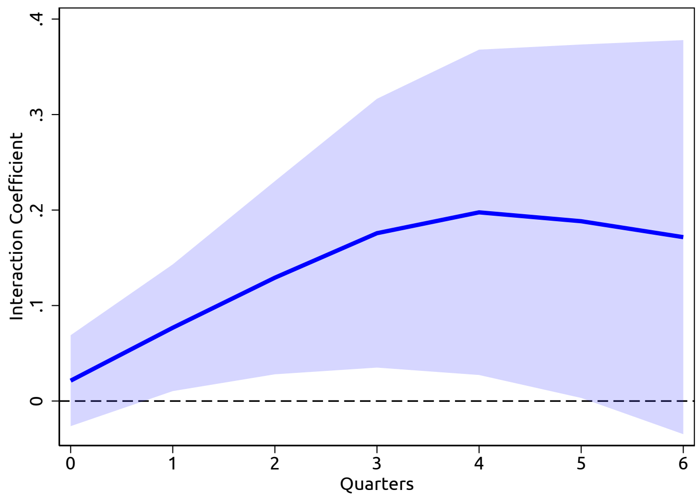
(c) Elasticity of vacancies to wages, $\hat{\beta}^{vw}$



(d) Elasticity of wages to wages, $\hat{\beta}^{ww}$

The figure plots the interaction coefficient between the monetary policy shock and equilibrium vacancy and wage measure in Equation (14). Please see caption of table 1 for more details.

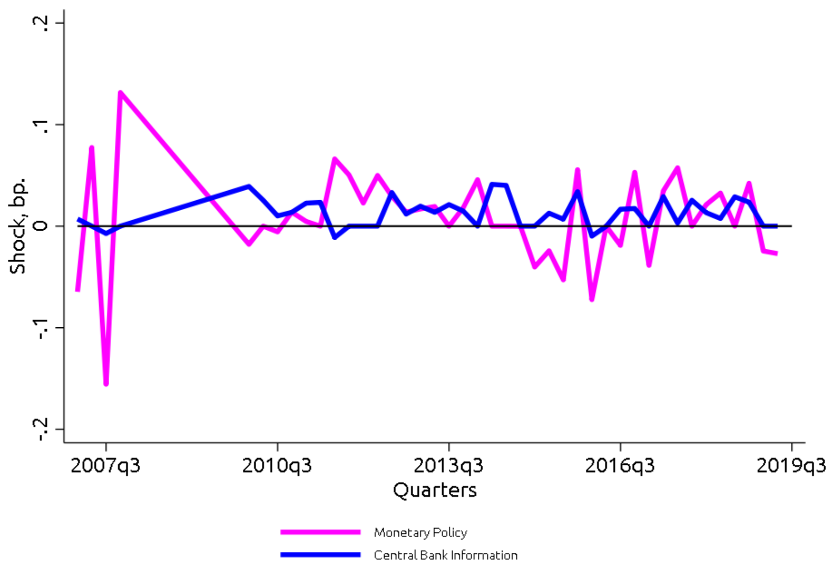
Figure 3: Interaction Coefficient of Labor Market Power and a Monetary Policy Easing Shock Across Horizons



This figure plots $\hat{\beta}_H$ from estimating $\sum_{h=0}^{H} \Delta \text{Employment}_{i,t+h} = \alpha_H + \beta_H \text{ MP easing}_t \times \text{Equilibrium vacancies}_{i,t-1} + \theta_H X_{i,t} + \gamma_{i,H} + \gamma_{t+H} + \varepsilon_{i,t+H}$ where $\Delta \text{Employment}_{i,t+h}$ is defined as the log difference between employment in time t+h and t-1. MP easing_t is the negative of the monetary policy shock by Jarociński and Karadi (2020), in which a positive value reflects monetary policy easing. $\text{LMP}_{i,t-1}$ is defined as one if the firm is in the top 5% of equilibrium vacancies in at least one commuting zone. For more details see subsection 4.4. Standard errors are clustered at the firm level. Shaded areas represent a 95% confidence interval.

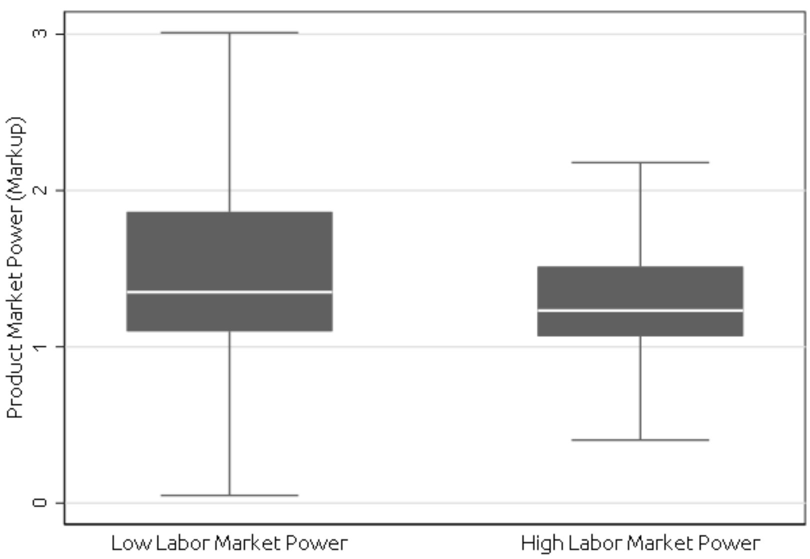
A Appendix Tables and Figures

Figure A1: Monetary Policy Shocks in Percentage Points (JK 2020)



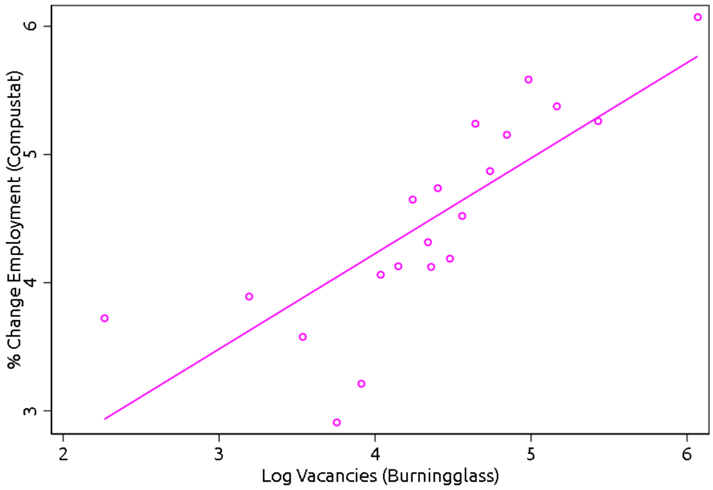
This figure plots Shocks time series used in the paper. The shock series is developed by Jarociński and Karadi (2020). The positive value of the shock reflects monetary policy easing. The unit is in percentage points. Monetary Policy reflects the shock component that can be assigned to the direct effects of Monetary Policy. Central Bank Information component measures the shock component associated with the effects of the Fed information.

Figure A2: Product Market Power and Labor Market Power



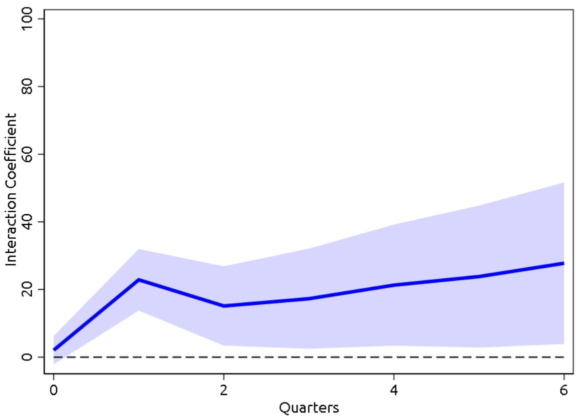
This figure plots the distribution of product market power for firms with and without labor market power. Labor market power firms are defined as those that are in the top 5% of labor market power in at least one commuting zone. Product market power is defined as the estimated markup following the approach by De Loecker et al. (2020). The white line reflects the median. The upper (lower) side of the box reflects the 75th (25th) percentile of the markup distribution. The adjacent lines at the top (bottom) reflect the maximum (minimum) value of the markup distribution.

Figure A3: Sensitivity of Employment to Vacancy Postings

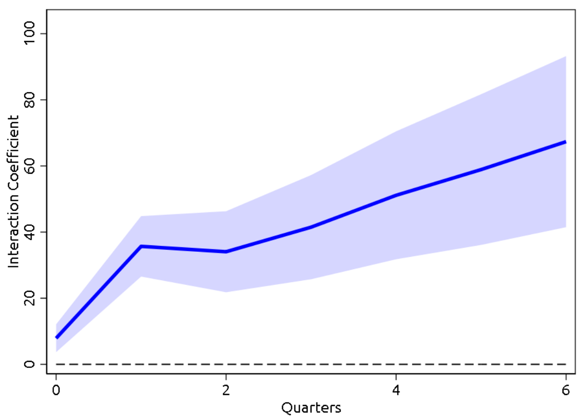


This figure plots a binscatterplot between the log change in employment from Compustat on Log Vacancy postings from Lightcast: $\Delta Employment_{i,t} = \alpha_i + \alpha_t + \beta_1 \text{Log Vacancies}_{i,t} + \epsilon_{i,t}$ where $\Delta Employment_{i,t}$ is the log change in employment of firm i between year t and $t-1$ in Compustat. $\text{Log Vacancies}_{i,t}$ is the log number of vacancies posted by firm i in year t from Lightcast.

Figure A4: Response of Vacancy Postings to a Monetary Policy Easing Shock Across Horizons Depending on Requirement of a College Degree



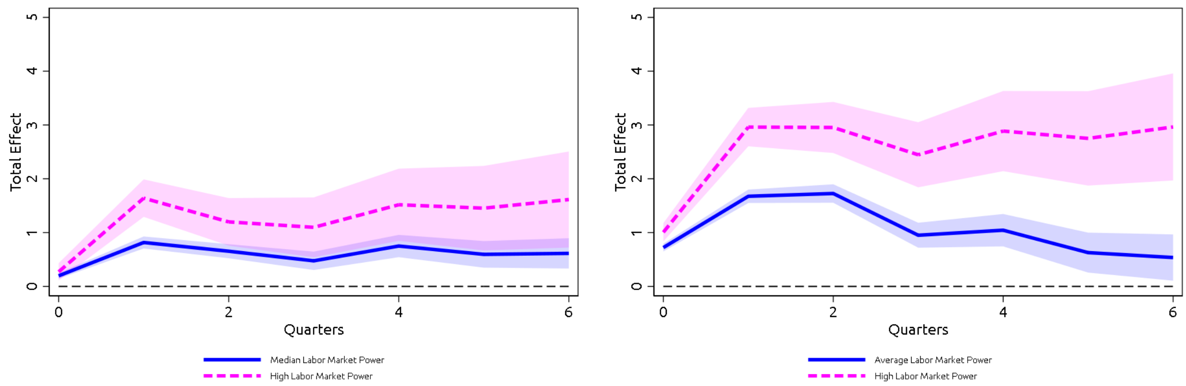
(a) Vacancies *with* college requirement



(b) Vacancies *without* college requirement

This figure plots $\hat{\beta}_H$ and $\hat{\beta}_H + \hat{\omega}_H$ from estimating $\sum_{h=0}^{H} \text{Log Vacancies}_{i,c,t+h} = \alpha_H + \beta_H \text{ MP easing}_t \times \text{LMP}_{i,c,t-1} + \omega_H \text{ MP easing}_t \times \text{LMP}_{i,c,t-1} \times \text{Vacancy Type} + \theta_H X_{i,c,t} + \gamma_{i,t+H} + \gamma_{c,t+H} + \varepsilon_{i,c,t+H}$ where $\text{Log Vacancies}_{i,c,t}$ is defined as the log number of vacancies posted by firm i, in commuting zone c in quarter t. Vacancy Type $= 1$ means that the vacancy requires a college degree. **The left panel** represents the response of the vacancies that require a college degree, **the right panel** represents vacancies that do *not* require a college degree. MP easing_t is the negative of the monetary policy shock by Jarociński and Karadi (2020), in which a positive value reflects monetary policy easing. $\text{LMP}_{i,c,t-1}$ is defined as the cumulative vacancy share of firm i in commuting zone c at quarter $t-1$. For more details see section 3. Standard errors are double clustered at the firm and commuting zone level. Shaded areas represent a 95% confidence interval.

Figure A5: Response of Vacancy Postings to a Monetary Policy Easing Shock Across Horizons Depending on Requirement of a College Degree and on Labor Market Power



(a) Vacancies *with* college requirement (b) Vacancies *without* college requirement

This figure plots the estimated response of vacancy postings for a firm a large extent of market power (95th percentile) in pink and medium market power (50th percentile) in blue from the following regression $\sum_{h=0}^{H}$ Log Vacancies$_{i,c,t+h}$ $= \alpha_H + \beta_H$ MP easing$_t \times$ LMP$_{i,c,t-1} + \omega_H$ MP easing$_t \times$ LMP$_{i,c,t-1} \times$ Vacancy Type $+ \theta_H X_{i,c,t} + \gamma_{i,t+H} + \gamma_{c,t+H} + \varepsilon_{i,c,t+H}$ where Log Vacancies$_{i,c,t}$ is defined as the log number of vacancies posted by firm i, in commuting zone c in quarter t. Vacancy Type = 1 means that the vacancy requires a college degree. **The left panel** represents the response of the vacancies that require a college degree, **the right panel** represents vacancies that do *not* require a college degree. MP easing$_t$ is the negative of the monetary policy shock by Jarociński and Karadi (2020), in which a positive value reflects monetary policy easing. LMP$_{i,c,t-1}$ is defined as the cumulative vacancy share of firm i in commuting zone c at quarter $t-1$. For more details see section 3. The pink lines are defined $\alpha + \beta \times$ LMP$_{i,c,t-1}(P95)$ for non-college vacancies and $\alpha + \beta \times$ LMP$_{i,c,t-1}(P95) + \omega \times$ LMP$_{i,c,t-1}(P95)$. The blue lines are defined as $\alpha + \beta \times$ LMP$_{i,c,t-1}(P50)$ and $\alpha + \beta \times$ LMP$_{i,c,t-1}(P50) + \omega \times$ LMP$_{i,c,t-1}(P50)$. Standard errors are double clustered at the firm and commuting zone level. Shaded areas represent a 95% confidence interval.

Table A1: Labor Demand Effect of Monetary Policy - Vacancies to Vacancies - Wage Sample

	Log Vacancies$_{i,c,t}$				
	(1)	(2)	(3)	(4)	(5)
MP easing$_t$	0.666***				
	(0.055)				
Equilibrium Vacancies$_{i,c,t}$	16.721***	12.330***	17.171***	19.714***	19.399***
	(1.703)	(1.648)	(1.733)	(1.727)	(1.915)
MP easing$_t$ × Equilibrium Vacancies$_{i,c,t}$	6.752***	6.181***	8.694***	10.048***	18.246***
	(2.504)	(2.336)	(2.607)	(2.663)	(4.894)
Obs.	3,546,366	3,546,366	3,546,366	3,545,581	2,715,673
Firm FE	✓	✓	✓	✓	
CZ FE	✓		✓		
Time FE		✓	✓		
CZ × Time FE				✓	✓
Firm × Time FE					✓
No. Firms	216,315	216,315	216,315	216,310	97,856

This table reports estimates of the following regression: Log Vacancies$_{i,c,t} = \alpha + \beta$ MP$_t \times$ Eq.Vac$_{i,c,t-1} + \theta Z_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where Log Vacancies$_{i,c,t}$ is defined as the log number of vacancies posted by firm i, in commuting zone c in quarter t. MP$_t$ is the negative of the monetary policy shock by Jarociński and Karadi (2020), and thus a positive value reflects monetary policy easing. Eq.Vac$_{i,c,t-1}$ is defined as the cumulative vacancy share of firm i in commuting zone c at quarter $t-1$. For more details see section 3. $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Table A2: Labor Demand Effect of Monetary Policy - Vacancies to Wages - Wage Sample

	Log Vacancies$_{i,c,t}$				
	(1)	(2)	(3)	(4)	(5)
MP easing$_t$	0.727***				
	(0.070)				
Equilibrium Wages$_{i,c,t}$	0.023***	-0.007	0.019***	0.019***	0.021***
	(0.007)	(0.008)	(0.007)	(0.007)	(0.007)
MP easing$_t$ × Equilibrium Wages$_{i,c,t}$	-0.110**	-0.121**	-0.087*	-0.086*	-0.030
	(0.050)	(0.048)	(0.047)	(0.046)	(0.048)
Obs.	1,828,640	1,828,641	1,828,640	1,827,037	1,359,325
Firm FE	✓	✓	✓	✓	
CZ FE	✓		✓		
Time FE		✓	✓		
CZ × Time FE				✓	✓
Firm × Time FE					✓
No. Firms	110,318	110,318	110,318	110,303	36,606

This table reports estimates of the following regression: Log Vacancies$_{i,c,t}$ $= \alpha + \beta$ MP$_t \times$Eq.Wage$_{i,c,t-1} + \theta Z_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where Log Vacancies$_{i,c,t}$ is defined as the log number of vacancies posted by firm i, in commuting zone c in quarter t. MP$_t$ is the negative of the monetary policy shock by Jarociński and Karadi (2020), and thus a positive value reflects monetary policy easing. Eq.Wage$_{i,c,t-1}$ is defined as the standartized average cumulative residual of the ONET-level regression. The ONET-level regression is given by Wage$_{i,c,n,t} = \alpha + \gamma_{n,t} + \gamma_{c,t}$, where Wage is the wage of the firm i in the commuting zone c at the time t for ONET occupation n; $\gamma_{n,t}$ are ONET-time fixed effects and $\gamma_{c,t}$ are CZ-time fixed effects. For more details see section 3. $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Table A3: Labor Demand Effect of Monetary Policy - ONET, Vacancies to Wages, no lag

	Log Vacancies$_{i,c,t}$				
	(1)	(2)	(3)	(4)	(5)
MP easing$_t$	1.156***	1.069***	1.058***		
	(0.345)	(0.345)	(0.347)		
Equilibrium Wages$_{i,c,t}$	0.136***	0.086***	0.084***	0.087***	0.085***
	(0.009)	(0.006)	(0.006)	(0.006)	(0.006)
MP easing$_t$ × Equilibrium Wages$_{i,c,t}$	-0.077**	-0.068**	-0.066**	-0.062*	-0.062*
	(0.031)	(0.031)	(0.032)	(0.032)	(0.033)
Obs.	17,711,407	17,711,407	17,707,982	17,711,060	17,707,574
Firm FE	✓	✓	✓	✓	✓
CZ FE	✓	✓			
ONET FE		✓		✓	
ONET × CZ FE			✓		✓
CZ × Time FE				✓	✓
No. Firms	252,173	252,173	252,148	252,172	252,147

This table reports estimates of the following regression: Log Vacancies$_{i,c,t}$ $= \alpha + \beta$ MP easing$_t \times$ X$_{i,c,n,t-1} + \theta Z_{i,c,n,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where Log Vacancies$_{i,c,t}$ is defined as the log number of vacancies posted by firm i, in commuting zone c in quarter t for the ONET-occupation n. MP easing$_t$ is the negative of the monetary policy shock by Jarociński and Karadi (2020), and thus a positive value reflects monetary policy easing. X$_{i,c,n,t-1}$ is defined as the log of the overtime average salary offered by firm i in commuting zone c at quarter $t-1$ for the ONET-occupation n. For more details see section 3. $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Table A4: Labor Demand Effect of Monetary Policy - ONET, Vacancies to Vacancies, no lag

	Log Vacancies$_{i,c,t}$				
	(1)	(2)	(3)	(4)	(5)
MP easing$_t$	0.450***	0.473***	0.471***		
	(0.028)	(0.028)	(0.027)		
Equilibrium Vacancies	0.774***	1.368***	1.770***	1.461***	1.912***
	(0.043)	(0.060)	(0.079)	(0.062)	(0.083)
MP easing$_t$ × Equilibrium Vacancies	0.443***	0.617***	0.811***	0.693***	0.906***
	(0.159)	(0.157)	(0.160)	(0.182)	(0.187)
Obs.	17,932,913	17,932,913	17,929,754	17,932,607	17,929,414
Firm FE	✓	✓	✓	✓	✓
CZ FE	✓	✓			
ONET FE		✓		✓	
ONET × CZ FE			✓		✓
CZ × Time FE				✓	✓
No. Firms	213,614	213,614	213,583	213,612	213,580

This table reports estimates of the following regression: Log Vacancies$_{i,c,n,t} = \alpha + \beta$ MP easing$_t \times \mathrm{X}_{i,c,n,t-1} + \theta Z_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where Log Vacancies$_{i,c,n,t}$ is defined as the log number of vacancies posted by firm i, in commuting zone c in quarter t for the ONET-occupation n. MP easing$_t$ is the negative of the monetary policy shock by Jarociński and Karadi (2020), and thus a positive value reflects monetary policy easing. $\mathrm{X}_{i,c,n,t-1}$ is defined as the cumulative vacancy share of firm i in commuting zone c at quarter $t-1$ for ONET-occupation n. For more details see section 3. $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Table A5: Labor Demand Effect of Monetary Policy - ONET, Wages to Wages, no lag

	Log Vacancies$_{i,c,t}$				
	(1)	(2)	(3)	(4)	(5)
MP easing$_t$	0.258	0.293	0.376		
	(0.407)	(0.404)	(0.401)		
Equilibrium Wages$_{i,c,t}$	0.934***	0.920***	0.922***	0.936***	0.937***
	(0.003)	(0.003)	(0.003)	(0.002)	(0.002)
MP easing$_t$ × Equilibrium Wages$_{i,c,t}$	-0.005	-0.008	-0.016	-0.002	-0.004
	(0.039)	(0.039)	(0.039)	(0.023)	(0.023)
Obs.	6,916,841	6,916,841	6,912,065	6,916,143	6,911,267
Firm FE	✓	✓	✓	✓	✓
CZ FE	✓	✓			
ONET FE		✓		✓	
ONET × CZ FE			✓		✓
CZ × Time FE				✓	✓
No. Firms	226,653	226,653	226,599	226,649	226,593

This table reports estimates of the following regression: Log Wages$_{i,c,t} = \alpha + \beta$ MP easing$_t \times \mathrm{X}_{i,c,n,t-1} + \theta Z_{i,c,n,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where Log Wages$_{i,c,t}$ is defined as the log deviation of the wage posted by firm i, in commuting zone c in quarter t for the ONET-occupation n from the regional average salary. MP easing$_t$ is the negative of the monetary policy shock by Jarociński and Karadi (2020), and thus a positive value reflects monetary policy easing. $\mathrm{X}_{i,c,n,t-1}$ is defined as the log of the overtime average salary offered by firm i in commuting zone c at quarter $t-1$ for the ONET-occupation n. For more details see section 3. $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Table A6: Labor Demand Effect of Monetary Policy - ONET, Wages to Vacancies, no lag

	Log Vacancies$_{i,c,t}$				
	(1)	(2)	(3)	(4)	(5)
MP easing$_t$	0.180***	0.233***	0.233***		
	(0.035)	(0.035)	(0.034)		
Equilibrium Vacancies	-0.107***	-0.042***	-0.032**	-0.001	0.029**
	(0.018)	(0.011)	(0.013)	(0.010)	(0.012)
MP easing$_t$ × Equilibrium Vacancies	0.114	0.011	-0.013	0.034	0.025
	(0.104)	(0.092)	(0.092)	(0.090)	(0.084)
Obs.	3,152,508	3,152,508	3,147,173	3,150,958	3,145,411
Firm FE	✓	✓	✓	✓	✓
CZ FE	✓	✓			
ONET FE		✓		✓	
ONET × CZ FE			✓		✓
CZ × Time FE				✓	✓
No. Firms	102,487	102,487	102,393	102,450	102,350

This table reports estimates of the following regression: Log Wages$_{i,c,t} = \alpha + \beta$ MP easing$_t \times \mathrm{X}_{i,c,n,t-1} + \theta Z_{i,c,n,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where Log Wages$_{i,c,t}$ is defined as the log deviation of the wage posted by firm i, in commuting zone c in quarter t for the ONET-occupation n from the regional average salary. MP easing$_t$ is the negative of the monetary policy shock by Jarociński and Karadi (2020), and thus a positive value reflects monetary policy easing. $\mathrm{X}_{i,c,n,t-1}$ is defined as the cumulative vacancy share of firm i in commuting zone c at quarter $t-1$ for ONET-occupation n. For more details see section 3. $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Table A7: Robustness to the Choice of Monetary Policy Shock

	Log Vacancies$_{i,c,t}$					
	(1)	(2)	(3)	(4)	(5)	(6)
MP easing$_t$ × Equilibrium Vacancies$_{i,c,t}$	6.014***	2.525***	-0.065***	0.059***	0.039***	0.082***
	(1.437)	(0.625)	(0.015)	(0.013)	(0.010)	(0.020)
R-squared	0.368	0.368	0.368	0.368	0.368	0.368
Obs.	15,791,442	15,791,442	15,069,930	15,069,930	15,069,930	15,069,930
Firm FE	✓	✓	✓	✓	✓	✓
CZ × Time FE	✓	✓	✓	✓	✓	✓
Shock	Nakamura and Steinsson (2018)	Bu et al. (2021)	Jarocinski (2021)	Jarocinski (2021)	Jarocinski (2021)	Jarocinski (2021)

This table reports estimates of the following regression: Log Vacancies$_{i,c,t} = \alpha + \beta$ MP easing$_t \times \mathrm{LMP}_{i,c,t-1} + \theta X_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where Log Vacancies$_{i,c,t}$ is defined as the log number of vacancies posted by firm i, in commuting zone c in quarter t. MP easing$_t$ are different monetary policy shocks, including Nakamura and Steinsson (2018), Bu et al. (2021) and Jarocinski (2021), in which a positive value reflects monetary policy easing. The four shocks from Jarocinski (2021) are: a standard contractionary monetary policy shock (column 3), a forward guidance shock that reflects a commitment future policy rates (column 4), a shock to longer-term treasury yields mostly affected by asset purchase announcements (column 5), and a forward guidance shock that captures the stance of future monetary policy in the sense of a prediction of the appropriate stance of policy (column 6). $\mathrm{LMP}_{i,c,t-1}$ is defined as the cumulative vacancy share of firm i in the corresponding commuting zone c at quarter $t-1$. For more details see section 3. $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

B Appendix Proofs and Propositions

B.1 Proposition 1

- *Under any classical monopsony model, there is a positive covariance between equilibrium vacancy shares and wages.*

Table A8: Robustness to the Choice of Monetary Policy Shock

	Log Vacancies$_{i,c,t}$					
	(1)	(2)	(3)	(4)	(5)	(6)
MP easing$_t$ × Equilibrium Wages$_{i,c,t}$	-0.025***	-0.005**	-0.0001	-0.00002	-0.0002**	0.0001
	(0.005)	(0.002)	(0.0001)	(0.00005)	(0.00007)	(0.0001)
R-squared	0.445	0.445	0.443	0.443	0.443	0.443
Obs.	4,454,151	4,454,151	4,135,037	4,135,037	4,135,037	4,135,037
Firm FE	✓	✓	✓	✓	✓	✓
CZ × Time FE	✓	✓	✓	✓	✓	✓
Shock	Nakamura and Steinsson (2018)	Bu et al. (2021)	Jarocinski (2021)	Jarocinski (2021)	Jarocinski (2021)	Jarocinski (2021)

This table reports estimates of the following regression: Log Vacancies$_{i,c,t}$ $= \alpha + \beta$ MP easing$_t$ × LMP$_{i,c,t-1}$ + $\theta X_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where Log Vacancies$_{i,c,t}$ is defined as the log number of vacancies posted by firm i, in commuting zone c in quarter t. MP easing$_t$ are different monetary policy shocks, including Nakamura and Steinsson (2018), Bu et al. (2021) and Jarocinski (2021), in which a positive value reflects monetary policy easing. The four shocks from Jarocinski (2021) are: a standard contractionary monetary policy shock (column 3), a forward guidance shock that reflects a commitment future policy rates (column 4), a shock to longer-term treasury yields mostly affected by asset purchase announcements (column 5), and a forward guidance shock that captures the stance of future monetary policy in the sense of a prediction of the appropriate stance of policy (column 6). LMP$_{i,c,t-1}$ is defined as the cumulative vacancy share of firm i in the corresponding commuting zone c at quarter $t-1$. For more details see section 3. $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

To prove the Proposition 1, we consider the relationship between equilibirum vacancies and wages. In the considered model $v_{ijt} = l_{ijt} = n_{ijt}$, hence, we obtain by definition:

$$\frac{\partial w}{\partial v}\frac{v}{w} = \frac{\partial w}{\partial n}\frac{n}{w} = \gamma$$

Table A9: Cumulative Labor Demand Effect of Monetary Policy across Vacancy Types, One Year Horizon

	$\sum_{h=0}^{3}$ Log Vacancies$_{i,c,t+h,j}$			
	(1)	(2)	(3)	(4)
LMP$_{i,c,t-1}$	60.537***	70.127***	57.072***	77.867***
	(4.516)	(5.117)	(4.418)	(5.653)
Type$_j$	-0.705***		-1.173***	
	(0.067)		(0.049)	
MP easing$_t$ × LMP$_{i,c,t-1}$	27.150**	38.581***	23.695**	39.483***
	(10.664)	(11.387)	(10.184)	(12.935)
MP easing$_t$ × Type$_j$	-0.467***		-0.463***	
	(0.128)		(0.123)	
LMP$_{i,c,t-1}$ × Type$_j$		-19.180***		-41.591***
		(3.412)		(3.610)
MP easing$_t$ × LMP$_{i,c,t-1}$ × Type$_j$		-22.864***		-31.576***
		(5.582)		(7.925)
Obs.	30,184,882	30,184,882	30,184,882	30,184,882
Vacancy Type	college	college	software	software
Firm × Time FE	✓	✓	✓	✓
CZ × Time FE	✓	✓	✓	✓
Vac. Type × Time FE		✓		✓

This table reports estimates of the following regression: $\sum_{h=0}^{3}$ Log Vacancies$_{i,c,t+h,j}$ $= \alpha + \beta$MP easing$_t$ × LMP$_{i,c,t-1}$ + δ MP easing$_t$ × LMP$_{i,c,t-1}$ × Type$_{i,c,t,j}$ $+ \theta X_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \gamma_{j,t} + \varepsilon_{i,c,t}$, where Log Vacancies$_{i,c,t,j}$ is defined as the log number of vacancies posted by firm i, in commuting zone c in quarter t. Vacancy Type$_{i,c,t,j} = 1$ for the type "college" means that vacancies require a college degree and for the type "software" means that vacancies require software skills. Note the sum includes 4 quarters and thus estimates reflect the effect over a year. MP easing$_t$ is the negative of the monetary policy shock by Jarociński and Karadi (2020), in which a positive value reflects monetary policy easing. LMP$_{i,c,t-1}$ is defined as the cumulative vacancy share of firm i in commuting zone c at quarter $t-1$. Type$_j$ is a dummy taking the value of one if the vacancy requires the particular "Vacancy Type" reported in the similar named column and zero otherwise. For more details see section 3. $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time, and $\gamma_{j,t}$ are vacancy type-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Extension. If other firms parameters are allowed to vary, the labor supply equation would instead linearize to:

$$\tilde{w}_{ij} = \frac{1}{\eta}\tilde{n}_{ij} + \left(\frac{1}{\theta} - \frac{1}{\eta}\right)\tilde{n}_j \qquad \tilde{n}_j = \sum \lambda_{ij}\tilde{n}_{ij}$$

Assuming that the changes in the parameters are iid across firms, the covariance becomes:

$$\begin{aligned}
&\text{cov}(\tilde{w}_{ij}, \tilde{s}_{ij}) = (1-s)\frac{1}{\eta}\text{var}(\tilde{n}_{ij}) + (1-s)\left(\frac{1}{\theta} - \frac{1}{\eta}\right)\text{cov}(\tilde{n}_{ij}, \tilde{n}_j) \\
&\text{with } \text{cov}(\tilde{n}_{ij}, \tilde{n}_j) = \theta^2\lambda\text{var}(\tilde{m}_{ij})
\end{aligned}$$

B.2 Proposition 2

- *Firms with greater search monopsony (larger α_{ij}) pay lower wages, but post* ***more*** *vacancies in equilibrium.*

- *There will be a negative covariance between vacancies and wages if search effects are sufficiently strong.*

To prove the first part of the Proposition 2, we consider the effect of the change in α:

$$\begin{aligned}
(w_{it})'_\alpha \frac{\alpha}{w} &= -\frac{1}{1-\alpha} \\
(v_{it})'_\alpha \frac{\alpha}{v} &= \frac{1}{1-\alpha}
\end{aligned}$$

The first equation is unambiguously negative, indicating that firms with larger matching elasticity pay more marked down wages as discussed in the text.

The second equation is positive for the average firm, given that $\frac{av}{n} = 1$ (meaning that matching intensity is calibrated in a way that it is equal to 1 for the average firm).

To prove the last part of the proposition 2, consider the formula for the covariance between vacancies and wages:

$$\begin{aligned}
&\text{cov}(\tilde{w}_{ij}, \tilde{s}_{ij}) = (1-s)\text{cov}(\tilde{w}_{ij}, \tilde{v}_{ij}) \\
&\text{cov}(\tilde{w}_{ij}, \tilde{v}_{ij}) = \frac{\alpha+\gamma}{1-\alpha}\text{cov}(\tilde{v}_{ij}, \tilde{l}_{ij}) - \frac{1+\gamma}{1-\alpha}\text{cov}(\tilde{v}_{ij}, \tilde{A}_{ij}) - \alpha\frac{1+\gamma}{1-\alpha}\text{var}(\tilde{v}_{ij})
\end{aligned}$$

$$
\begin{aligned}
\operatorname{cov}(\tilde{l}_{ij}, \tilde{v}_{ij}) &= \frac{\alpha}{(1-\alpha)^2}\operatorname{var}(\tilde{\alpha}_{ij}) + \frac{1}{(1-\alpha)^2}\left(1+\frac{1}{\gamma^*}\right)^2 \operatorname{var}(\tilde{A}_{ij}) \\
&+ \frac{1}{(1-\alpha)^2}\left(1+\frac{1}{\gamma^*}\right)\left(\alpha+\frac{1}{\gamma^*}\right)\operatorname{var}(\tilde{m}_{ij}) + \frac{\alpha}{(1-\alpha)^2}\left(1+\frac{1}{\gamma^*}\right)\left(1+\frac{\alpha}{\gamma^*}\right)\operatorname{var}(\tilde{c}_{ij}) \\
\operatorname{cov}(\tilde{A}_{ij}, \tilde{v}_{ij}) &= \frac{1}{(1-\alpha)}\left(1+\frac{1}{\gamma^*}\right)\operatorname{var}(\tilde{A}_{ij}) \\
\operatorname{var}(\tilde{v}_{ij}) &= \frac{1}{(1-\alpha)^2}\operatorname{var}(\tilde{\alpha}_{ij}) + \frac{1}{(1-\alpha)^2}\left(1+\frac{1}{\gamma^*}\right)^2\operatorname{var}(\tilde{m}_{ij}) + \frac{1}{(1-\alpha)^2}\left(1+\frac{\alpha}{\gamma^*}\right)^2\operatorname{var}(\tilde{c}_{ij}) \\
&+ \frac{1}{(1-\alpha)^2}\left(1+\frac{1}{\gamma^*}\right)^2\operatorname{var}(\tilde{A}_{ij})
\end{aligned}
$$

where $\gamma^* = \gamma\left(1+\frac{\psi}{1+\gamma}\right)$ The same formula would arise present for Kimball substitution between labor markets.

Extension. If other firms parameters are allowed to vary, the labor supply equation would instead linearize to:

$$
\tilde{w}_{ij} = \frac{1}{\eta}\tilde{n}_{ij} + \left(\frac{1}{\theta}-\frac{1}{\eta}\right)\tilde{n}_j - \tilde{A}_{ij} - \alpha(\tilde{v}_{ij}-\tilde{n}_{ij}) \qquad \tilde{n}_j = \sum \lambda_{ij}\tilde{n}_{ij}
$$

Assuming that changes in firm-level parameters are iid, the covariance becomes:

$$
\begin{aligned}
&\operatorname{cov}(\tilde{w}_{ij}, \tilde{s}_{ij}) = (1-s)\operatorname{cov}(\tilde{w}_{ij}, \tilde{v}_{ij}) \\
&\operatorname{cov}(\tilde{w}_{ij}, \tilde{v}_{ij}) = \frac{1}{\eta}\operatorname{cov}(\tilde{v}_{ij}, \tilde{n}_{ij}) + \left(\frac{1}{\theta}-\frac{1}{\eta}\right)\operatorname{cov}(\tilde{v}_{ij}, \tilde{n}_j) - \alpha\operatorname{cov}\left(\tilde{v}_{ij}, \left(\frac{\tilde{v}}{n}\right)_{ij}\right) - \operatorname{cov}(\tilde{v}_{ij}, \tilde{A}_{ij}) \\
&\text{with } \operatorname{cov}(\tilde{v}_{ij}, \tilde{n}_{ij}) = \frac{k}{(1-\alpha)^2}\operatorname{var}(\tilde{m}_{ij}) + \frac{\alpha k}{(1-\alpha)^2}\operatorname{var}(\tilde{c}_{ij}) + \frac{k}{(1-\alpha)^2}\operatorname{var}(\tilde{A}_{ij}) + \operatorname{var}(\tilde{n}_{ij}) \\
&\qquad \operatorname{cov}(\tilde{v}_{ij}, \tilde{n}_j) = \frac{\lambda\theta(1+\theta)}{(1-\alpha)^2}\operatorname{var}(\tilde{m}_{ij}) + \frac{\lambda\theta(1+\theta)}{(1-\alpha)^2}\operatorname{var}(\tilde{A}_{ij}) + \frac{\lambda\theta(1+\theta)\alpha^2}{(1-\alpha)^2}\operatorname{var}(\tilde{c}_{ij}) \\
&\qquad \operatorname{cov}\left(\tilde{v}_{ij}, \left(\frac{\tilde{v}}{n}\right)_{ij}\right) = \frac{1+k}{(1-\alpha)^2}\operatorname{var}(\tilde{m}_{ij}) + \frac{1+k}{(1-\alpha)^2}\operatorname{var}(\tilde{A}_{ij}) + \frac{1+k\alpha^2}{(1-\alpha)^2}\operatorname{var}(\tilde{c}_{ij}) + \frac{1}{(1-\alpha)^2}\operatorname{var}(\tilde{\alpha}_{ij}) \\
&\qquad \operatorname{cov}(\tilde{v}_{ij}, \tilde{A}_{ij}) = \frac{(1+k)}{(1-\alpha)}\operatorname{var}(\tilde{A}_{ij}) \\
&k = \left(\theta\lambda + \frac{1}{\frac{1}{\eta}+\frac{\gamma\psi}{\gamma+1}}(1-\lambda)\right) \qquad \psi = \frac{\partial\gamma}{\partial n}\frac{n}{\gamma} = \left(\frac{1}{\theta}-\frac{1}{\eta}\right)\left(\frac{1}{\eta}+1\right)\frac{\lambda}{\gamma}
\end{aligned}
$$

B.3 Proposition 3

In a model without search monopsony and following a monetary policy shock, if firms with varying degrees of classical monopsony power do not change their wages differentially, vacancies of these firms also do not change differentially. However, if firms have varying *degrees of search monopsony power and don't change their wages differentially, vacancies of firms with greater search monopsony power react more.*

First consider a case where firms have a γ that does not depend on the market shares, but differs across firms. In this case, labor supply equation will be updated to:

$$\left(\frac{a_i v_i}{n_i}\right)^{\alpha_i} \frac{w_i}{WMD} = \left(\frac{n_i}{N}\right)^{\gamma_i} (1+\gamma_i)$$

where $D_t^{-1} = \int_0^1 (1+\gamma_i) \left(\frac{n_{it}}{N_t}\right)^{1+\gamma_i} di$ Monetary policy response will be given by:

$$\hat{v}_i = \frac{1}{1-\alpha_i}\hat{m}_i + \frac{1}{\gamma_i}\left(\frac{\alpha_i}{1-\alpha_i}\hat{m}_i + \hat{w}_i - \frac{\int \lambda_i \frac{1}{\gamma_i}\left(\frac{\alpha_i}{1-\alpha_i}\hat{m}_i + \hat{w}_i\right) di}{\int \lambda_i \frac{1}{\gamma_i} di}\right) + \hat{N}$$

where λ_i is the equilibrium payroll share, $\lambda_i = \frac{w_i l_i}{WL}$.

If wages respond homogeneously as assumed in the proposition, $\hat{w}_i = \hat{w}$, then we can take them out of the integral in the numerator of the term on the right, which then is simply $-\hat{w}$ and can be canceled out with $+\hat{w}$ that just preceeds it. Equation (7) then simplifies to:

$$\hat{v}_i = \frac{1}{1-\alpha_i}\hat{m}_i + \frac{1}{\gamma_i}\left(\frac{\alpha_i}{1-\alpha_i}\hat{m}_i - \frac{\int \lambda_i \frac{1}{\gamma_i} \frac{\alpha_i}{1-\alpha_i}\hat{m}_i di}{\int \lambda_i \frac{1}{\gamma_i} di}\right) + \hat{N}$$

First, consider the case when firms only differ in the extent of their classical monoposony power, or γ_i but $\alpha_i = \alpha$ and $\hat{m}_i = \hat{m}$. In this case, vacancies would respond homogeneously, just like wages.

The second part of Proposition 3 can be proven by considering firms with the same γ and $\hat{m}$ but different α_i. Assuming wages do not respond differentially, the response of vacancies would depend positively on α_i.

In the baseline model, the response is given in the main text, and, similarly, there will be no heterogeneity if there is no heterogeneity in α_i. A similar equation will appear if parameters η and θ are allowed to vary across firms.

The same result would persist in a more general case.

Extension 1: *In the absence of heterogeneity in search monopsony and of heterogeneity in productivity or any other non-monopsony factor that would generate heterogeneous responses of wages to monetary policy and assuming that the substitution between labor markets is homogeneous of degree one such that γ_i only depends on the own search share $\frac{n_i}{N}$, heterogeneity in classical monopsony alone cannot generate an heterogeneous response of vacancies.*

Examples of this type of substitution include Kimball.

Under this extension, the monetary policy response is still the same and given by:

$$\hat{v}_i = \frac{1}{1-\alpha_i}\hat{m}_i + \frac{1}{\gamma_i}\left(\frac{\alpha_i}{1-\alpha_i}\hat{m}_i + \hat{w}_i - \frac{\int \lambda_i \frac{1}{\gamma_i}\left(\frac{\alpha_i}{1-\alpha_i}\hat{m}_i + \hat{w}_i\right)di}{\int \lambda_i \frac{1}{\gamma_i}di}\right) + \hat{N} \tag{18}$$

where λ_i is the equilibrium payroll share, $\lambda_i = \frac{w_i l_i}{WL}$.

Under this extension, the zero hypothesis for no homogeneity in α_i remains the same: if matching efficiency is heterogeneous, the response of vacancies is heterogeneous as well. If vacancies respond heterogeneously to monetary policy, so that firms with higher equilibrium vacancies and lower equilibrium wages respond more – there must be heterogeneity in α_i across firms.

Before considering the extension, we prove the following lemma. Note that, for simplicity this lemma is formulated from the perspective of a consumer problem, but, as we will see in the main proof, a similar structure would emerge in the context of the model that we consider here.

Lemma 1 For homogeneous of degree one substitution between markets and if "natural pricing assumption holds", the log-deviations of quantities and wages (quantities and prices) are such that:

$$\int \frac{p_i y_i}{PY}\hat{y}_i di = \hat{Y}$$
$$\int \frac{p_i y_i}{PY}\hat{p}_i di = \hat{P}$$

From natural pricing assumption have:

$$PY = \int_i p_i y_i di$$

From homogeneous of degree one substitution, the aggregation equation is given by the following functional:

$$F\left(\frac{y_i}{Y}\right) = 1$$

Assume additionally that this functional is both Gateaux and Fréchet differentiable.

Conduct a simple cost minimization to derive first-order conditions (λ is a Lagrange multiplier):

$$\int p\psi_i di - \lambda dF\left(\frac{y_i}{Y};\frac{\psi_i}{Y}\right) = 0 \quad \forall \psi(i) \in C^\infty$$

$$dF\left(\frac{y_i}{Y};\frac{\psi_i}{Y}\right) = \frac{\int p\psi_i di}{\lambda}$$

Substitute $\psi_i = y_i$ and use the natural pricing assumption to obtain:

$$dF\left(\frac{y_i}{Y};\frac{y_i}{Y}\right) = \frac{PY}{\lambda}$$

Now consider the aggregation equation and some deviation $\hat{y}_i = \frac{\psi_i}{y_i}$ and $\hat{Y} = \frac{\Psi}{Y}$ from a point where cost minimization holds. Substitute the first-order conditions from the cost-minimization problem.

$$dF\left(\frac{y_i}{Y};\frac{\psi_i}{Y}\right) - dF\left(\frac{y_i}{Y};\frac{y_i}{Y^2}\Psi\right) = 0$$

$$\frac{\int p\psi_i di}{\lambda} - \frac{PY}{\lambda}\frac{\Psi}{Y} = 0$$
$$\frac{PY\int \frac{p_i y_i}{PY}\frac{\psi_i}{y_i}di}{\lambda} - \frac{PY}{\lambda}\frac{\Psi}{Y} = 0$$

$$\hat{Y} = \int \frac{p_i y_i}{PY}\hat{y}_i di$$

Now use the natural pricing and the result for quantities to derive an equation for prices:

$$\hat{P} + \hat{Y} = \int \frac{p_i y_i}{PY}(\hat{p}_i + \hat{y}_i)di$$

$$\hat{P} = \int \frac{p_i y_i}{PY}\hat{p}_i di$$

End of Lemma 1

Coming back to the problem at hand, consider the labor income maximization problem:

$$WL = \max_{n_i} \int w_i \left(\frac{v_i}{n_i'}\right)^{\alpha} n_i di$$
$$F\left(\frac{n_i}{N}\right) = 1$$

Where F is a homogeneous of degree one differentiable functional, representing the substitution between labor markets.

Lemma 1 can be applied in this problem after a substitution:

$$\bar{W}N = \max_{n_i} \int \bar{w}_i n_i di$$
$$F\left(\frac{n_i}{N}\right) = 1$$

where $\bar{W} = W\frac{L}{N}$ and $\bar{w}_i = w_i\left(\frac{v_i}{n_i'}\right)^{\alpha}$

Therefore, the deviations of the aggregate search efforts N can be expressed as:

$$N = \int \frac{\bar{w}_i n_i}{\bar{W}N} \hat{n}_i di = \int \frac{w_i l_i}{LW} \hat{n}_i di = \int \lambda_i \hat{n}_i di$$

As in the baseline model, firm's first-order conditions can be expressed as:

$$w_{it} = \frac{1-\alpha_i}{1+\gamma_{it}} m_{it}$$
$$v_{it} = \alpha_i \frac{l_{it}}{c} m_{it}$$

where $\frac{1}{\gamma_{it}} = \frac{\partial n_{it}}{\partial \bar{w}_{it}} \frac{\bar{w}_{it}}{n_{it}}$

Now consider the response to monetary policy (change in the marginal revenue from labor m):

$$\hat{v}_i - \hat{n}_i = \frac{1}{1-\alpha_i}\hat{m}$$

where $\eta_i = \frac{\partial \gamma_i}{\partial \frac{n_i}{N}} \frac{n_i}{N} \frac{1}{\gamma_i}$.

Note the difference with the standard case: $\eta \neq 0$, indicating non-zero superelasticity (η itself is not superelasticity, but can be expressed in terms of it).

Consider the labor supply equation and the result of Lemma 1 to obtain:

$$\hat{n} = \frac{1}{\gamma^*}\left(\frac{\alpha_i}{1-\alpha_i}\hat{m} + w_i - \frac{\int \lambda_i \frac{1}{\gamma_i^*}\left(\frac{\alpha_i}{1-\alpha_i}\hat{m}_i + w_i\right) di}{\int \lambda_i \frac{1}{\gamma_i^*} di}\right) + \hat{N}$$

Substitute into the optimal solution for vacancy postings to obtain:

$$\hat{v}_i = \frac{1}{1-\alpha_i}\hat{m}_i + \frac{1}{\gamma_i^*}\left(\frac{\alpha_i}{1-\alpha_i}\hat{m}_i + \hat{w}_i - \frac{\int \lambda_i \frac{1}{\gamma_i^*}\left(\frac{\alpha_i}{1-\alpha_i}\hat{m}_i + \hat{w}_i\right) di}{\int \lambda_i \frac{1}{\gamma_i^*} di}\right) + \hat{N}$$

Note that if there is no heterogeneity in α_i, there is no homogeneity in response of either vacancies or wages.

B.4 Extension 2

Extension 2: *Under decreasing returns to scale or imperfect competition in the product market; if, following a monetary policy shock, firms with varying degrees of classical monopsony power do not change their wages differentially, vacancies of these firms also do not change differentially. However, under the same conditions, vacancies of firms with greater search monopsony power react more.*

Decreasing returns to scale will result in $\hat{m}$ depending on the level of employment within the firm. So the response of vacancies and wages under flexible wages will become:

$$\hat{w}_i = \hat{m}_i = \hat{m}p_i - \kappa_i l_i$$

Under Cobb-Douglas production function with decreasing returns to scale $y = zl^a$, $\kappa_i = (1-a)$. Under imperfect competition in the product markets, $\kappa_i = \frac{1}{\xi^d}$ with ξ^d being elasiticity of demand for products of the firm. Note that in both cases κ_i is between zero and one.

In both those cases, the response of vacancies to monetary policy becomes:

$$\hat{v}_i = \frac{1}{1-(1-\kappa_i)\alpha_i}\hat{m}_i + \frac{(1-\kappa_i)(1-\alpha_i)}{1-(1-\kappa_i)\alpha_i}\frac{1}{\gamma_i}\left(\frac{\alpha_i}{1-\alpha_i}\hat{m}_i + \hat{w}_i - \frac{\int \lambda_i \frac{1}{\gamma_i}\left(\frac{\alpha_i}{1-\alpha_i}\hat{m}_i + \hat{w}_i\right) di}{\int \lambda_i \frac{1}{\gamma_i} di}\right) + \hat{N}$$

As in the proposition 3, if there is no heterogeneity in wage response and only heterogeneity in γ_i, there will be no heterogeneity in vacancy response.

B.5 Extension 3

Under wage rigidity, if, following a monetary policy shock, firms with varying degrees of classical monopsony power do not change their wages differentially, vacancies of these firms also do not change differentially. However, under the same conditions, vacancies of firms with greater search monopsony power react more.

Under wage rigidity, the formula for the response of vacancy postings does not change:

$$\hat{v}_i = \frac{1}{1-\alpha_i}\hat{m}_i + \frac{1}{\gamma_i}\left(\frac{\alpha_i}{1-\alpha_i}\hat{m}_i + \hat{w}_i - \frac{\int \lambda_i \frac{1}{\gamma_i}\left(\frac{\alpha_i}{1-\alpha_i}\hat{m}_i + \hat{w}_i\right) di}{\int \lambda_i \frac{1}{\gamma_i} di}\right) + \hat{N} \qquad (19)$$

where λ_i is the equilibrium payroll share, $\lambda_i = \frac{w_i l_i}{WL}$

First order conditions and the response of wages change, however, if there is still no differential wage response to monetary policy, there will be no differential response of vacancies to monetary policy in the model with only classical monopsony power. One example for the particular assumptions about wage rigidity is considered below.

Assume Calvo wage setting, so that each period wages are only adjusted with probability θ.

In this model, the first-order conditions change and are listed below. For the wage the optimal reset wage at time t is given

$$w^*_{it} = \frac{\sum_{\tau \geq t} \theta^{\tau - t} l_{i\tau} \xi^w_{i\tau} m_{i\tau}}{\sum_{\tau \geq t} l_{i\tau}(\xi^w_{i\tau} + 1)}$$

$$v_{it} = \xi^v_{it} \frac{l_{it}}{c} m_{it}$$

Log-linearization of the optimal response to monetary policy shock is given by:

$$\hat{w}_{it} = \theta(1-\theta)\sum_{\tau \geq t} \theta^{\tau - t}\left[\hat{m}_{i\tau} + \frac{-\gamma_i \eta_i}{\gamma_i + 1}(\hat{n}_{i\tau} - \hat{N}_t)\right] + (1-\theta)\hat{w}_{it-1}$$

$$\hat{v}_i = \frac{1}{1-\alpha_i}\hat{m}_i + \frac{1}{\gamma_i}\left(\frac{\alpha_i}{1-\alpha_i}\hat{m}_i + \hat{w}_i - \frac{\int \lambda_i \frac{1}{\gamma_i}\left(\frac{\alpha_i}{1-\alpha_i}\hat{m}_i + \hat{w}_i\right) di}{\int \lambda_i \frac{1}{\gamma_i} di}\right) + \hat{N}$$

where $\eta = \frac{\partial \gamma}{\partial \frac{n}{N}} \frac{n/N}{\gamma}$, λ_i is the equilibrium payroll share, $\lambda_i = \frac{w_i l_i}{WL}$